Wildfire and Landscape Change

Paul M Santi[a], and Francis K Rengers[b], [a] Colorado School of Mines, Golden, CO, United States and [b] U.S. Geological Survey, Golden, CO, United States

© 2020.

***Change History*: October 2020. PM Santi and FK Rengers updated the text and added recent references, updated figures 2, 4, 13, 18, and 19, and added new figure 10, and several aspects of the text were updated to reflect recent research discoveries.**

1 Introduction

Wildfire can act as an agent of landscape change by removing the protection provided by plant life and enabling the forces of water and wind and temperature variations to more freely act on the landscape surface (**Shakesby and Doerr, 2006**; **Moody and Martin, 2009**; **Moody et al., 2013**). Wildfire can also impart physical changes to soil structure (**Badía and Martí, 2003**; **Nyman et al., 2013**), increasing its susceptibility to downslope movement. Fine ash and accumulations of water-repellent plant oils can change runoff and infiltration patterns (**DeBano et al., 1979**; **Larsen et al., 2009**). With rainfall, these changes enhance overland flow and erosion rates. This "fire-flood" sequence was first recognized in regions with Mediterranean climates such as southern Europe and southern California (**Kotok and Kraebel, 1935**), but also occurs throughout the world in fire-susceptible areas with steep slopes, abundant sediment supplies, and sufficient rainfall. Larger mass movements such as debris flows and landslides can result from the increased runoff and sediment mobilization and from loss of root strength (e.g., **Cannon, 2001**; **Meyer et al., 2001**) (**Fig. 1**). Cumulative effects of fire over time on landforms include hillslope denudation (**Robichaud et al., 2010**; **Wagenbrenner et al., 2010**; **Rengers et al., 2016b**), incision and aggradation of gullies and stream channels (**DeLong et al., 2018**; **Nyman et al., 2019**), and building of alluvial and debris fans (**Fig. 2**) (**Meyer et al., 1992**; **Moody and Martin, 2001**).

The temperature at which fires burn and the duration of heating can be considered the primary controlling factor for fire-induced changes (**Fitzpatrick, 1980**; **Chandler et al., 1983**; **Ryan and Noste, 1985**; **Ulery et al., 1996**; **Arocena and Opio, 2003**; **Moody et al., 2005**). Several variables influence both the temperature that wildfires generate and how long a part of the landscape is subjected to those temperatures. Peak wildfire temperatures are strongly dependent on fuel availability and spatial distribution, whether it is live or dead, its moisture content, and the vegetation species burned (**Ryan and Noste, 1985**; **Ryan, 2002**; **Wohlgemuth et al., 2006**). Fire temperatures and residence times are also affected by the topography the fire passes through and prevailing weather conditions, including wind speed, relative humidity, and ambient temperature. Vegetation on steep hillslopes tends to burn more completely than that on ridgetops or in flat meadows, and the hottest wildfire temperatures have been measured in topographic chutes lined with heavy vegetation (**Goudie et al., 1992**). In California, wildfires burning in grass and lower montane forests typically generate temperatures between 210 °C and 250 °C. Higher temperatures up to 700 °C are recorded for fires that consume shrublands or slash piles (**Wohlgemuth et al., 2006**). Peak temperatures reported from many different settings throughout the world range up to 1100 °C (**Goudie et al., 1992**). Those peak soil surface temperatures are subsequently propagated downward, altering soil properties (Campbell et al., 1994, 1995; **Massman and Frank, 2005**; **Massman, 2012**; **Rengers et al., 2017**).

Burn severity, a metric used to characterize the effects of fire temperature and duration on above-ground vegetation and soil organic matter (**Ryan and Noste, 1985**; **Keeley, 2009**), affects the intensity of runoff and erosion in a landscape. Although burn severity magnitudes are defined using several different methods, the definitions usually focus on the size and extent of the fuels consumed and the effects of the fire on the surface soil (**Keeley, 2009**). **Ryan and Noste (1985)** define the highest burn severity as areas where canopy trees are killed, and surface litter and soil organic layers are largely consumed. Moderate burn severity results in the mortality of some canopy trees, understory plants that are charred or consumed, and the pre-fire organic layer largely consumed. Low burn severity is characterized by green foliage remaining on canopy trees, charred or consumed surface litter, and a largely intact soil organic layer. Researchers explored potential quantitative relations between soil hydraulic parameters, which govern post-fire runoff rates, and burn severity magnitudes, and found broad trends associating higher burn severity with lower infiltration (**Moody et al., 2016**). Moreover, areas burned at the highest severities respond to rainfall with the highest runoff and erosion rates (**Benavides-Solario and MacDonald, 2005**). However, debris flows are most likely to be produced from basins burned at a combination of high and moderate severities (**Staley et al., 2017**), and areas burned at moderate and low severities can also experience increases in overland flow and soil losses, as well as severe erosion (**Shakesby and Doerr, 2006**; **Parson et al., 2010**). For example, **Gartner et al. (2008)** defined a statistical relation between debris-flow volume and, among other factors, the total area of basin burned at a combination of high, moderate, and low severities. Similarly, **Pelletier and Orem (2014)** found that sediment yield was primarily controlled by slope and burn severity.

The type of landscape exposed to disturbance by wildfire directly influences its sensitivity to change. Landscapes with high relief and erodible materials show strong responses due to greater vegetative variability, geomorphic processes that are more sensitive to vegetative disturbance, and more topographic influence on fire behavior (**Swanson, 1981**). Landscapes with low relief and erosion-resistant bedrock show fewer and lower magnitude changes in geomorphic processes following wildfire (**Swanson, 1981**).

Although large parts of Earth, including a wide variety of landscapes, experience both human-caused and natural fires on a seasonal basis (**Fig. 3**), the frequency and severity of wildfires have changed over time in many parts of the world in response to both changing climatic conditions and land-management practices (**Steele et al., 1986**; **Barrett, 1988**; **Meyer et al., 1995**; **Grissino-Mayer et al., 2004**). For example, charcoal from sediment cores in the Oregon Coast Range showed increasing fire intervals from about 110 years in 7000–4850 BCE,

UNCORRECTED PROOF

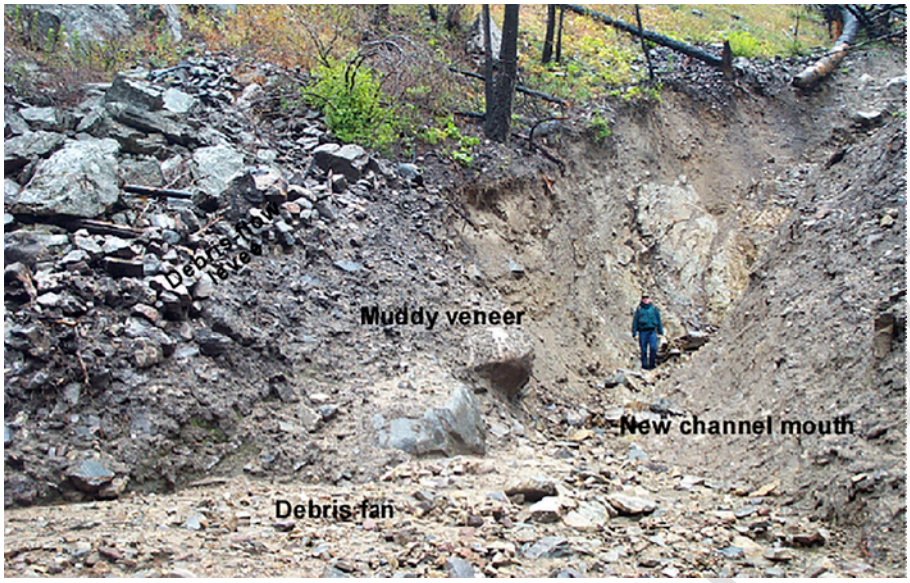


Fig. 1 Severe downcutting and bank failure from a debris flow following a wildfire near Sula, Montana. Debris-flow deposits from this event occur in the fan at the freshly incised channel mouth and at levees and muddy veneers on the hillslopes adjacent to the channel. Photograph by Susan Cannon, U.S. Geological Survey.

to about 160 years in 4850–750 BCE, to about 230 years in 750 BCE to present, as climatic conditions generally became cooler and more humid (**Long et al., 1998**). Charcoal derived from wildfire is often linked to severe drought periods in the southwestern United States (**Fitch and Meyer, 2016**). In fire-prone areas such as the western United States, modern fire suppression techniques have also led to fuel buildup in some forested regions (**Murphy et al., 2018**). Continued climate change is expected to lead to increases in the number of wildfires and their extents, and in the lengths of typical wildfire seasons in several western US ecosystems (**Stocks et al., 1998**; **Westerling et al., 2006**; **Backlund et al., 2008**) with the magnitude of these changes varying with ecosystem (Littell et al., 2009). Projections of increased precipitation associated with climate change (**Wentz et al., 2007**) would indicate the possibility of increasingly intensified impacts of fire on the landscape.

In this article, we examine physical changes brought about by wildfire in terms of temperature-induced physical losses and gains of materials and changes in soil, rock, and hydrologic properties. We then consider how these changes affect hillslope and channel erosion, transport, and deposition processes, and how these changes might ultimately be reflected within the landscape. This discussion is followed by examples of the use of geomorphic analyses and mapping to assess post-fire conditions and potential debris-flow hazards.

2 Physical changes brought about by wildfire

2.1 Physical losses

Depending on its severity, wildfire can remove some or all of the vegetation and litter cover, thereby altering key variables in the hydrologic cycle (**Shakesby and Doerr, 2006**). Full or partial consumption of the tree canopy, shrubs, and herbaceous plants can temporarily reduce, or stop completely, both rainfall interception and evapotranspiration (**Tiedemann et al., 1979**; **Díaz-Fierros et al., 1994**; **Loaiciga et al., 2001**; **Atchley et al., 2018**; **Poon and Kinoshita, 2018**). For example, **Hanshaw et al. (2009)** found that rain gages located beneath a chaparral shrub canopy received an average of 42% less rainfall than did gages located in the adjacent burned area, and 1-h rainfall intensities were reduced by up to 67%. **Klock and Helvey (1976)**, **Helvey (1980)**, **Megahan (1983)**, and Cannon et al. (1998) attributed changes in soil moisture conditions observed after wildfires to decreased evapotranspiration rates brought about by fire-induced vegetation mortality. Consumption of soil-mantling litter and duff can also reduce rainfall storage rates, and the loss of organic matter can decrease the water retention of soils (**Ebel, 2012**). With removal of surface litter, water storage capacities decreased from pre-fire conditions in Portugal by approximately 0.5 mm for each centimeter of litter burned in pine forests, and by 3 mm for each centimeter of litter burned in a commercial eucalypt stand (**Shakesby and Doerr, 2006**).

Removal of vegetation stems that buttress loose pebbles and soil on hillslopes and consumption of the network of fine roots and soil-binding organisms within surficial soils can lead to increased dry-ravel production and downslope movement of up to cobble-sized materials, either during or immediately after fires (**Rice, 1982**; **Wells, 1987**; **Florsheim et al., 1991**; **Gabet, 2003**; **Roering and Gerber, 2005**; **Shakesby and Doerr, 2006**; **Lamb et al., 2011**; **DiBiase and Lamb, 2013**) (**Fig. 4**; see also **Section 3.5** for additional discussion of fire-related dry-ravel processes).

Loss of surface cover leaves soil grains susceptible to dislodging by raindrop impact and subsequent entrainment by overland flow (**McGuire et al., 2016**) and can also result in changes in soil moisture dynamics (**Atchley et al., 2018**) with increased solar heating (**Shakesby**

UNCORRECTED PROOF



	Vegetation	Soils	Rock	Hydrology
Physical Changes	Blowdown Post-fire falls Short-term shallow root loss Long-term deep root decay Removal of litter and duff Needle cast	Reduced organic matter Ash addition Hydrophobic layer development Lower strength Decreased water content Particle aggradation Aggregate dispersion	Spalling Cracking	Change in storage capacity Decreased interception and infiltration Increased raindrop impact
Process Changes	Reduced evapotranspiration	Erosion Runoff Debris flows Landslides Dry ravel	Rockfall Talus	Increased overland flow Increased rilling and gullying Increased peak streamflow
Landform Changes	Hillslope erosion Channel aggradation Channel incision Debris / Alluvial fan development			

Fig. 2 Summary of geomorphic impacts of wildfire. Modified from Swanson FJ (1981) Fire and geomorphic processes. In: Mooney HA, Bonniksen TH, Christensen NL, Lotan JE, and Reiners WA (eds.), *Fire Regimes and Ecosystem Properties*, pp. 401–420. USDA Forest

Service General Technical Report WO-26.

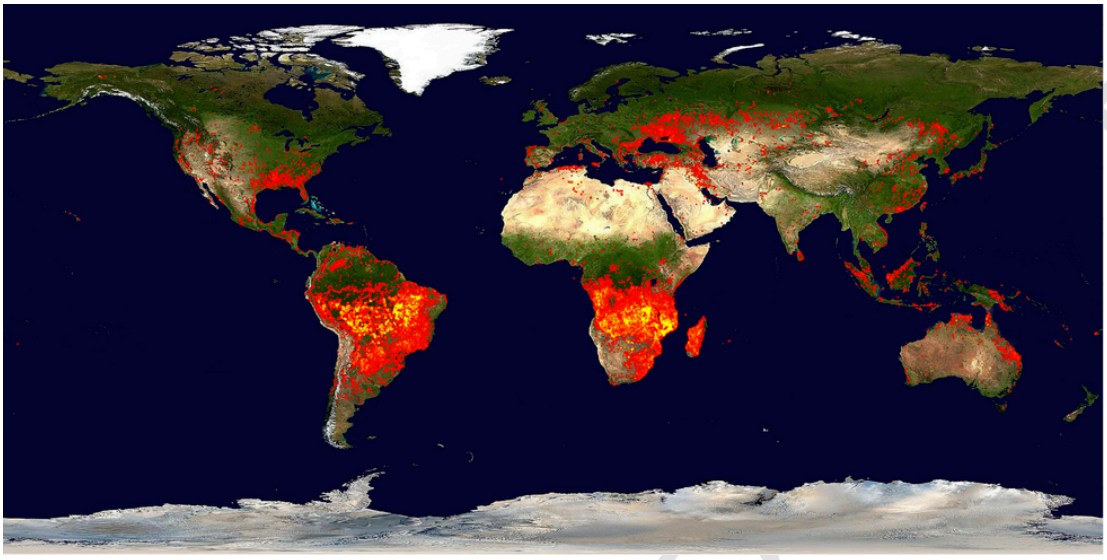


Fig. 3 Example of NASA's Fire Information for Resource Management System (FIRMS) map. Satellite data are plotted within 3 h. Note the widespread distribution of wildfires worldwide. Red colors indicate areas mapped by the VIIRS satellite and yellow colors indicate areas mapped by the Modis satellite. Image by NASA.



Fig. 4 Exposed slope within the perimeter of the 2016 San Gabriel Complex fire. An apron of sandy dry ravel has accumulated at the base of the slope on a forest road. Photograph by Francis Rengers, U.S. Geological Survey.

UNCORRECTED PROOF

and Doerr, 2006). Exposure of the bare soil also allows mineral or ash particles to move into soil pores; moisture-induced sealing by mineral or ash particles has been shown to enhance post-fire sediment yields (**Durgin, 1985**; **Neary et al., 2005**; **Larsen et al., 2009**; see **Section 3.1** for additional discussion of the effects of this phenomenon on post-fire geomorphic processes).

Within the top few centimeters of soil, consumption of organic matter, including fine roots, soil microbes, and mycorrhizal fungi can cause a loss or degradation of aggregate stability, which leads to increases in the ease of detachment-driven erosion (**Díaz-Fierros et al., 1994**; **Molina et al., 1994**; **Neary et al., 1999**; **Shakesby and Doerr, 2006**; **Mataix-Solera et al., 2009**; **Nyman et al., 2013**). Soil microbial and fungal activity are thought to contribute to aggregate stability through the secretion of cohesive compounds and the production of stabilizing fungal hyphae (**DeBano et al., 1979**; **Mataix-Solera et al., 2009**).

The physical process of heating during a fire can also remove soil moisture from surficial soils (**Moody and Martin, 2009**; **Moody and Ebel, 2012**). In addition to decreasing soil moisture to below the field capacity or driving off the soil moisture that exists within soil pore spaces, temperatures during fires can remove even the tightly bound film of capillary water that coats soil particles (**Bachmann and van der Ploeg, 2002**). Depletion of capillary water will inhibit subsequent movement of water into the soil (**Scott, 1963**), which is contrary to traditional infiltration theory that would predict the highest levels of infiltration during the driest state (**Ebel and Moody, 2013**). **Puri and Asghar (1940)** found that the hygroscopicity **(**the ability of particles to attract and hold water molecules) of some soils decreased with increases in temperatures above 400 °C. These soils adsorbed only about 0.01 cm^3/cm^3 of water at 10% humidity, and humidity may be even lower during a fire (**Moody and Martin, 2009**). Soil moisture measured for samples subjected to 550 °C in the laboratory had values less than 0.001 cm^3/cm^3, and samples taken immediately following a wildfire had soil moisture values as low as 1.0×10^{-7} cm^3/cm^3 (**Moody and Ebel, 2012**). Thus, after a fire the physical process of re-adsorption and reforming of capillary water must occur before any significant re-wetting or filling of pore spaces within the soil can occur. The time lag in wetting the soil above the field capacity may partially explain why disproportionately high runoff rates have been observed during the first rainstorms to impact an area before the re-wetting process is complete (**Moody and Martin, 2009**; Moody and Ebel, 2012).

Over longer time periods, wildfire-induced vegetation mortality can potentially lead to decreases in the strength properties of hillslope soils due to the decay of regolith-anchoring roots, and instability effects may appear months or years after the fire. Roots provide a tensile strength to soil, which can be considered as an additional cohesion term in stability models (**Schmidt et al., 2001**). This additional term can be much higher than the cohesion of either the soil or fractured rock mass alone (**Abe and Ziemer, 1991**). **Swanson (1981)** proposed that root decay could result in decreased apparent soil cohesion and, thus, an increased probability of shallow landsliding after fires. Studies of the effects of timber harvesting have shown decreases in root strength over time (**Burroughs and Thomas, 1977**; **Schmidt et al., 2001**), with the strongest effect 4–10 years after tree mortality (**Clayton and Megahan, 1986**) (**Fig. 5**). The effects on slope stability of increased soil moisture and decreased cohesion that follow timber harvesting have been studied extensively, leading to the conclusion that steep hillsides are commonly only marginally stable and that changes in this delicate equilibrium, such as loss of a component of the mass strength, can lead to slope failure (**Sidle, 2005**). More rapid declines in root strength following fire than those documented after timber harvest have been observed, suggesting that root strength was either compromised prior to the fire or that the heat of the fire damaged roots in the shallow subsurface (**Jackson and Roering, 2009**) (**Fig. 6**). In addition, several studies have observed landslides in areas that burned five or more years

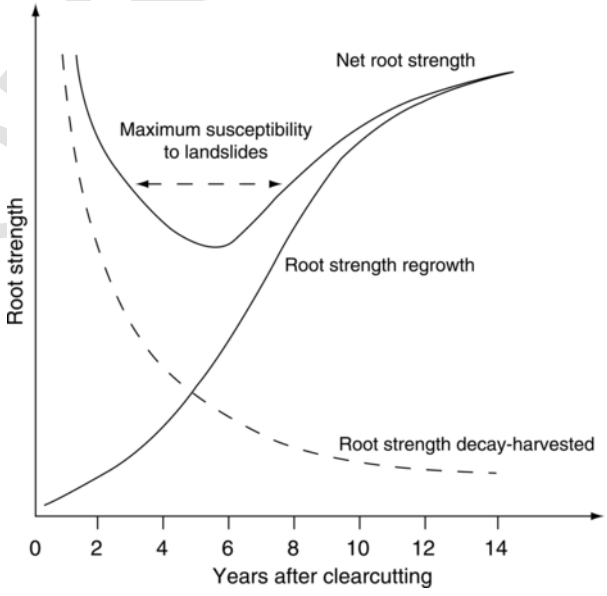


Fig. 5 An example of typical changes in forest vegetation rooting strength after timber harvesting. Root decay and recovery curves are based on numerous data worldwide compiled by Sidle (1991, 1992). Reproduced from Sidle RC (2005) Chapter 16: Influence of forest harvesting activities on debris avalanches and flows. In: Jakob M and Hungr O (eds.), *Debris-flow Hazards and Related Phenomena,* pp. 387–409. Berlin: Praxis/ Springer.

UNCORRECTED PROOF

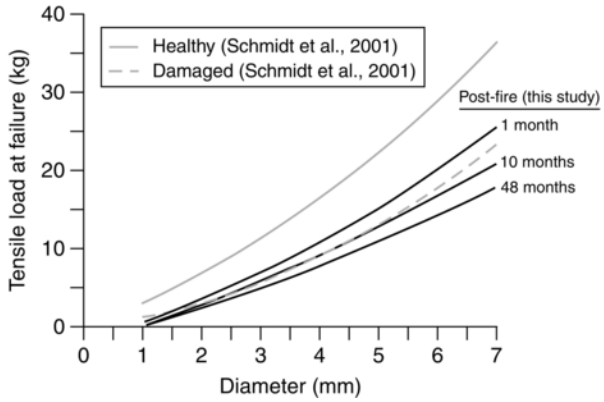


Fig. 6 Data from **Schmidt et al. (2001)** showing decrease in root strength over time following wildfire. Thick line shows strength of healthy roots, dashed line shows strength of damaged roots, and thin lines show strengths at three times after the fire. Reproduced from Jackson M and Roering JJ (2009) Post-fire geomorphic response in steep, forested landscapes: Oregon Coast Range, USA. *Quaternary Science Reviews* 29: 1131–1146. doi:10.1016/j.quasscirev.2008.05.003.

beforehand (**Rice and Foggin III, 1971**; **Benda and Dunne, 1997**; **Meyer et al., 2001**; **May and Gresswell, 2003**), and the landslides can transition to debris flows (**Jackson and Roering, 2009**).

Although fire is most frequently discussed as a way to develop or enhance existing soil-water repellency, passage of a fire can also reduce or completely destroy water repellency in a soil (Figs. 7 and 8) (discussed in detail in the following section). Passage of a fire through permafrost can also result in thawing of the soil ice (**Swanson, 1981**; **Minsley et al., 2016**; **Ebel et al., 2019**).

2.2 Physical gains

After a fire, a layer of fine ash and charcoal formed by the combustion of vegetation and the litter and duff layers may mantle the landscape (**Shakesby and Doerr, 2006**; **Balfour, 2007**; **Woods and Balfour, 2008**) (**Fig. 7**). The amount of ash and its physical and chemical characteristics depend on the type and quantity of fuels present, the degree of combustion, which varies with fuel density and moisture content, and fire severity (**Ulery et al., 1993**; **Neary et al., 2005**). Low-severity fires typically leave a thin (< 1 cm) layer of black ash



Fig. 7 Wetted black ash layer overlying dry water-repellent soil. Keys indicate the scale. Photograph by Charles Luce, U.S. Department of Agriculture Forest Service.

UNCORRECTED PROOF

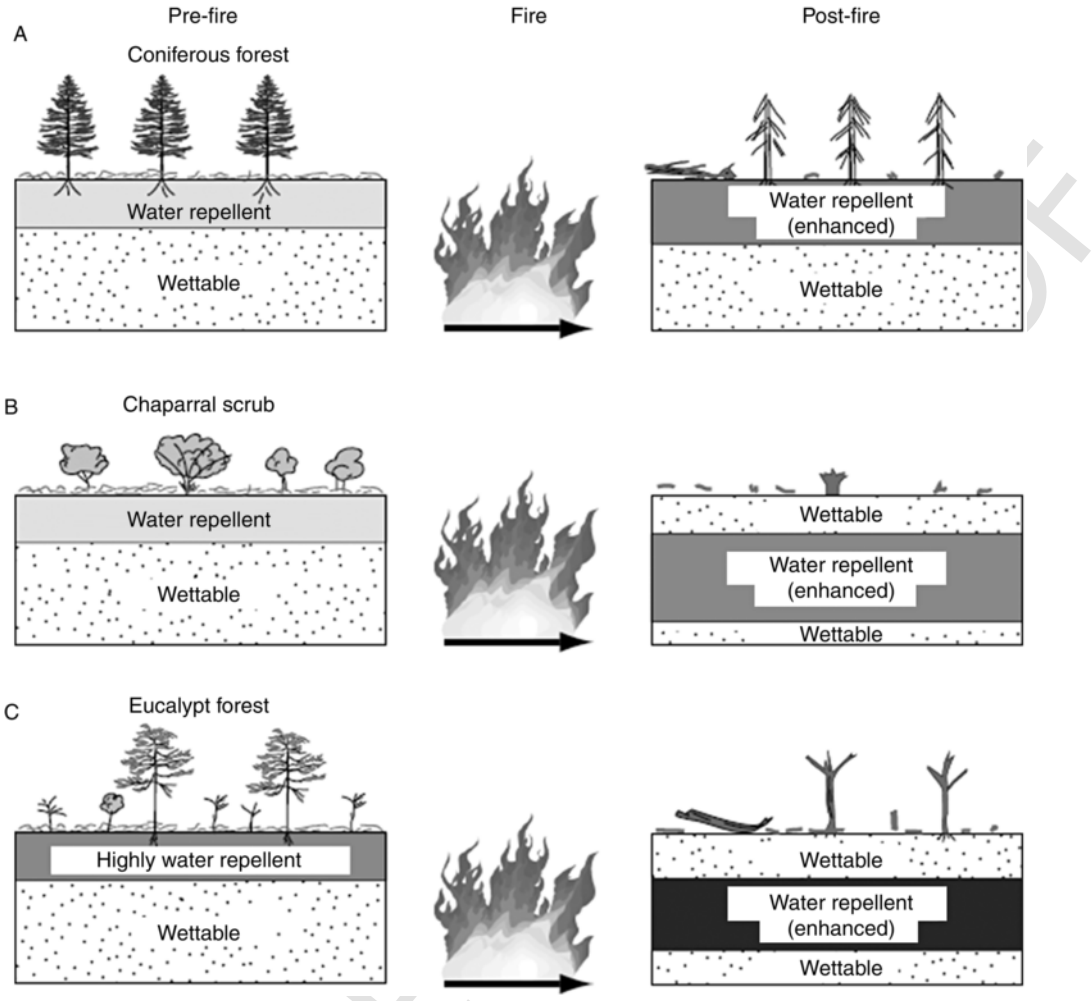


Fig. 8 Different soil wetting scenarios attributed to burning. (A) Water repellency is enhanced at or near the soil surface after burning. This effect has been observed after moderate and high severity fires in coniferous forests in the western US (**Huffman et al., 2001**; **Woods et al., 2007**). (B and C) High surface temperatures destroy surface water repellency while creating increased repellency a few centimeters below the surface. This phenomenon has been reported for hot chaparral fires in the western United States by DeBano (2000) and for eucalypt fires in Australia by Doerr et al. (2009). Darker shading represents more severe repellency. Reproduced from Doerr SH, Shakesby RA, and MacDonald LH (2009) Soil water repellency—A key factor in post-fire erosion. In: Cerdà A, and Robichaud PR (eds.), *Restoration Strategies After Forest Fires,* pp. 197–223. Enfield, NH: Science Publishers.

containing mostly char particles (**Woods and Balfour, 2008**), whereas high-severity fires can result in up to 20 cm of fine, predominantly black and white ash consisting mostly of carbonates and oxides of metals and silica, with small amounts of phosphorous, sulfur and nitrogen (**Ulery et al., 1993**; **Goforth et al., 2005**; **Gabet and Sternberg, 2008**). Sealing of soil pore spaces by ash particles has been shown to enhance post-fire sediment yields (**Larsen et al., 2009**). However, these effects are poorly understood, as laboratory analyses of ash effects on a pure sand suggest that ash clogging did not directly lead to changes in soil infiltration due to the high hydraulic conductivity of ash (**Stoof et al., 2016**). Ash has been shown to have hydrophilic properties (**Woods and Balfour, 2010**), which can buffer an incoming rainstorm from the underlying soil (**Ebel, 2012**). In some settings, water from precipitation can form a hard crust on the ash, which can further inhibit infiltration (**Balfour et al., 2014**); however, in many settings, high winds can remove wildfire ash from a site on a scale of tens of days (**Pereira et al., 2013**). See **Section 3.5** for additional discussion of the effects of this phenomenon on post-fire geomorphic processes.

A frequently cited change to hydrologic properties associated with fire is the development or enhancement of soil-water repellency (Figs. 7 and 8). Soil-water repellency refers to the inability of water to wet or infiltrate a dry soil (**Doerr et al., 2000**; **Dekker et al., 2005**). This phenomenon has been documented in a wide range of vegetation types and climates, including both unburned and burned terrains (Doerr et al., 2000, 2009). **DeBano (2000)** described water repellency developing during burning when hydrophobic organic substances in the litter and surficial soil are volatilized. Volatilization is thought to create a pressure gradient in the heated soil layer that causes some material to travel downward toward lower pressures. Condensation of this material then occurs at cooler temperatures at depth (**DeBano, 2000**). Heat generated by fire is also thought to improve bonding of hydrophobic substances to soil particles (**Savage et al., 1972**) and to make them more water

UNCORRECTED PROOF

repellent through pyrolysis (the thermochemical decomposition of organic material in the absence of oxygen) (**Giovannini, 1994**) and by changing the structural arrangement of the hydrophobic compounds (**Doerr et al., 2009**). In addition, high heat imparted by a wildfire can cause hyper-dry conditions (**Moody and Ebel, 2012**) immediately after a fire, which reduces the ability of capillary suction to infiltrate incoming water.

The intensity and persistence of soil-water repellency will differ with fire temperature and duration, vegetation type, soil moisture and texture, and time since burning (**Huffman et al., 2001**). Although fire can impart water repellency to a previously non-water-repellent soil, it can also either enhance or destroy a preexisting water-repellent condition. While higher severity fires typically generate stronger water repellency (**Tiedemann et al., 1979**; **DeBano, 2000**; **Huffman et al., 2001**), extremely hot fires can temporarily destroy the natural water repellence of soils (**Doerr and Moody, 2004**; **Doerr et al., 2009**) (**Fig. 8**). Laboratory experiments indicate that preexisting water repellency will be intensified at temperatures between 175 °C and 270 °C but will be destroyed at temperatures between 270 °C and 400 °C (**DeBano et al., 1976**; **Robichaud and Hungerford, 2000**; **Doerr et al., 2004**). These changes are independent of soil type, but depend on heating durations, with longer heating durations decreasing the temperatures at which these changes occur (**DeBano, 2000**; **Robichaud and Hungerford, 2000**; **Doerr et al., 2004**).

Vegetation type affects the amount and type of hydrophobic compounds available for translocation in a soil and thus the degree of water repellency (**DeBano, 2000**). Soils under vegetation with oil- or wax-rich leaves, such as sclerophyllous shrubs, conifers, and eucalypts are much more likely to develop strong water repellency than those under broad-leaved deciduous forests (**Doerr et al., 2000**).

Coarse-textured soils are commonly more water repellent for a given fire severity than finer textured soils. For instance, **Crockford et al. (1991)** documented an increase in water repellency with particle size. The difference has been attributed to the much smaller particle surface areas per unit volume of coarser soils. It is reasoned that, given a limited supply of hydrophobic particles to coat soil particles, it is more likely that the organic molecules will obtain complete coverage in a coarse soil than in a fine soil (**Meeuwig, 1971**; **DeBano, 2000**; **Doerr et al., 2009**). However, strong water repellency has also been documented in fine-grained soils (**Doerr et al., 1996**; **Robichaud and Hungerford, 2000**).

Because fire severity, vegetation cover, soil texture and antecedent soil moisture all differ across a burned area, the spatial distribution, depth, and thickness of soil water repellency also differs considerably (**Robichaud and Miller, 1999**; **Doerr and Moody, 2004**; **Woods et al., 2007**). The spatial distributions of water-repellent soils can range from isolated patches, to discrete layers with discontinuities (**Dekker and Ritsema, 1996**; **Jackson and Roering, 2009**), to continuous layers (**Woods et al., 2007**). The different infiltration rates between isolated patches can lead to preferential flow in macropores, which can account for 70–90% of the flow (**Nyman et al., 2010**), and the macropore distribution tends to increase with time after fire (**Nyman et al., 2014**). Recent models have shown that it is important to account for the variability in the spatial distribution of infiltration rates (**Nyman et al., 2014**). However, soil-water infiltration parameters, such as effective saturated hydraulic conductivity, vary with the measurement technique and the scale of the hydrologic measurements (**Langhans et al., 2016**) and are hard to measure accurately.

Soil-water repellency can be temporarily reduced during wet weather periods following a wildfire and then reappear as the soils dry (**Nyman et al., 2014**), but the effects of fire-induced soil-water repellency decrease over time as the hydrophobic compounds in the soil break down. The time for the water repellency to reach background levels ranges from as little as a few weeks to as much as 6 years (**DeByle, 1973**; **Dyrness, 1976**; **DeBano et al., 1998**; **MacDonald and Huffman, 2004**). Regional comparisons of hydrologic recovery after wildfire show that recovery after wildfires doesn't follow a standard trajectory (**Ebel and Martin, 2017**). Soil water repellency can disappear once the soil moisture content exceeds a threshold value that is specific to soil texture and, in some cases, organic matter content (**Doerr and Thomas, 2000**; **Dekker et al., 2001**; **Doerr et al., 2003**; **MacDonald and Huffman, 2004**; **Benavides-Solario and MacDonald, 2005**). Soil water repellency may also reappear, either gradually or immediately, as a wetted soil dries out (**Shakesby et al., 1993**; **Doerr et al., 2009**). The physical processes driving this response are not well understood.

2.3 Changes in hydrologic properties

In addition to changes in rainfall interception, evapotranspiration, and storage that occur in response to wildfires discussed above, other hydrologic properties that change significantly in response to wildfire include rainfall infiltration and runoff, and the critical shear stress for initiation of particle motion.

A number of studies have documented decreases in infiltration in response to wildfire, and comparison of infiltration rates measured from both unburned and burned soils is shown in **Fig. 9**. Water infiltration into soils is thought to be controlled by a gravitational component (characterized by a field-saturated hydraulic conductivity parameter Kfs) and a capillary suction component (characterized by a sorptivity parameter S), and while both of these measures are found to decrease in burned soils, the largest changes have been observed in the sorptivity (**Ebel and Moody, 2017**). Although responses vary, infiltration rates in burned soils are typically 20–80% less than those in similar unburned soils. Reductions in infiltration rates after fires have also been tied to observations of compaction from raindrop impact (**Swanson, 1981**; **Singer and Le Bissonnais, 1998**; **Martin and Moody, 2001**; **Wagenbrenner et al., 2006**; **Doerr et al., 2009**), changes in bulk density during heating (**Stoof et al., 2010**), blocking of soil pores by ash and fine material (**Larsen et al., 2009**), and hyper-dry conditions (**Moody and Ebel, 2012**). **Martin and Moody (2001)** tested infiltration rates at mountainous sites in New Mexico and Colorado, and showed that geology did not affect infiltration rates of burned soils, but after burning, infiltration rates differed strongly with the dominant pre-burn vegetation. They attribute this difference in part to the variability in litter and duff cover and in part to ash accumulation associated with each vegetation type. Variability in infiltration rates may also reflect different measurement techniques, soil types, and burn severity and soil moisture conditions (**Swanson, 1981**; **Kinner and Moody, 2008**; **Langhans et al., 2016**; **Ebel et al., 2016**). Despite this variability, defining an effective saturated hydraulic conductivity for a watershed based on field measurements is possible (**McGuire et al., 2018**).

Because of reduced infiltration rates, rainfall on areas burned by wildfires may lead to overland flow. In contrast to forested catchments where it is rarely observed, overland flow is common on burned terrain (**Shakesby and Doerr, 2006**). **Soler et al. (1994)** documented burned

UNCORRECTED PROOF

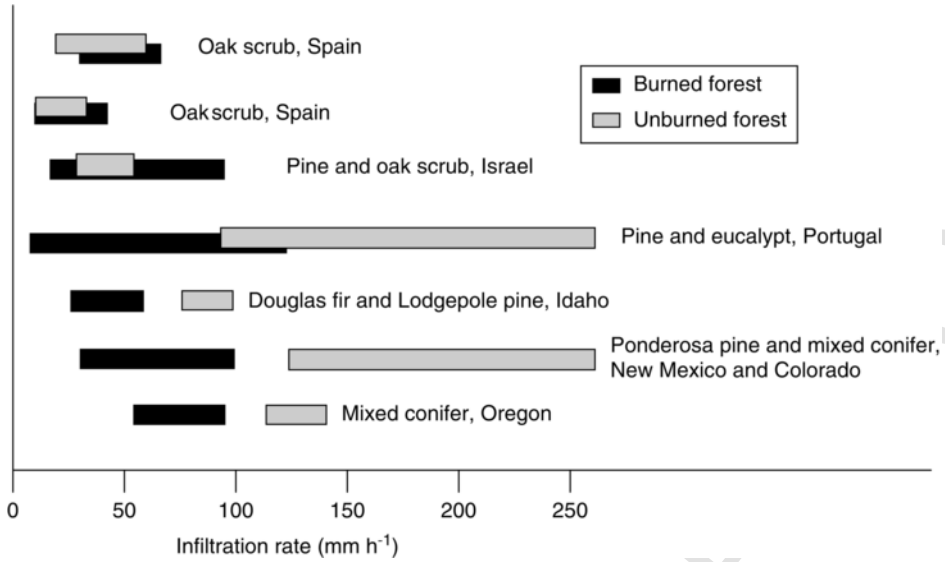


Fig. 9 Examples of measured ranges of soil infiltration rates in burned and unburned areas.

overland flow rates near Barcelona that were 14.6 times unburned rates, whereas **Prosser (1990)** measured overland flow on burned plots at 1.5–3 times the rates on unburned plots in New South Wales, Australia. **DeBano et al. (1998)** found that if vegetation and litter are reduced by fire to less than 10% surface cover, runoff can increase by more than 70%. Moreover, as burned areas recover, the rate of recovery, influenced by site conditions such as solar radiation, can influence runoff several years after fire, with arid watersheds continuing to produce more runoff than moist watersheds (**Noske et al., 2016**). The key factors governing the differences in runoff during wildfire recovery are changes to water repellency and surface roughness (**Noske et al., 2016**).

2.4 Changes in soil mineralogy and physical properties, and rock structure

Changes to the mineralogical character of soils in response to temperatures of 400 °C or greater are reported by **Humphreys et al. (2003)** and **Certini (2005)** (**Table 1; Ketterings et al., 2000**). These effects are confined to the upper few centimeters of soil, and they primarily impact iron oxides and clays. **Certini (2005)** suggested that the lack of a wildfire impact on other minerals may be because they require a minimum temperature of 500 °C for their dehydroxylation, the required initial stage in the heat-induced disruption of many minerals. In the case of iron oxides and clays, the principal changes include thermal transformation of goethite into maghemite and destruction of kaolinite. Heat-induced clay dispersion has also been described by **Durgin (1985)** and **Mills and Fey (2004)**.

Physical properties of soils, including the particle-size distribution, aggregate stability, bulk density, and elasticity can also change during burning. **Hubbert et al. (2002)** excavated pits into coarse loamy soils following a prescribed burn in southern California and by comparing burned areas to unburned areas, found that the average clay content decreased from 5.4% to 3.4%. Exposure to high temperatures during wildfires has also been shown to result in decreases in both clay- and silt-size fractions and a corresponding increase in the sand-size range. **Giovannini et al. (1988)** documented that at temperatures near 220 °C, organometallic cements binding soil particles are destroyed. Heating

Table 1 Fire-induced changes in soil minerals.

Mineral	*Temperature*	*Result*
Goethite, maghemite, and magnetite	> 400 °C	Conversion to very fine magnetite and/or and maghemite
Kaolinite	> 550 °C	Crystal lattice destroyed or converted to a poorly crystalline aluminosilicate cementing agent
Gibbsite	> 600 °C	Reduced concentration
Chlorite, chlorite-vermiculite, vermiculite, and hydroxy-interlayered vermiculite	600–700 °C	Transformation to illite

Compiled from Ketterings QM, Bigham JM, and Laperche V (2000) Changes in soil mineralogy and texture caused by slash-and-burn fires in Sumatra, Indonesia. *Soil Science Society of America Journal* 64: 1108–1117; Humphreys GS, Shakeby RA, Doerr SH, Blake WH, Wallbrink P, and Hart DM (2003) Some effects of fire on the regolith. In: Roach IC (ed.), *Advances in Regolith,* pp. 216–220. Canberra, Australia: CRC, LEME, and Certini G (2005) Effects of fire on properties of forest soils—A review. *Oecologia* 143: 1–10.

UNCORRECTED PROOF

to above 460 °C drives off hydroxyl groups from clays and results in structure loss. Such decreases in fine material, breaking down of cements, and loss of bonds lead to less cohesive soils that are easily eroded (**Durgin, 1985**; **Giovannini et al., 1988**; **DeBano et al., 1998**; **Mills and Fey, 2004**). In some cases, high temperatures are reported to fuse soil particles, creating coarser, but low-cohesion soil aggregates that are vulnerable to erosion (**DeBano et al., 1998**; **Neary et al., 1999**; **Wondzell and King, 2003**; **Blake et al., 2007**; **Riebe, 2008**; **Mataix-Solera et al., 2011**). An example of this fusion at the microscopic level is shown in **Fig. 10**. However, there is also evidence that when high temperatures fuse soil particles, the resultant aggregates are more stable and exhibit a corresponding decrease in erodibility (**Mataix-Solera and Doerr, 2004**). Soil-aggregate stability can also decrease due to the presence of ash; in a series of laboratory experiments, **Durgin (1985)** demonstrated that ash, when mixed with water, creates a high pH solution that facilitates dispersion of soil aggregates. This loss has been documented to correspond to decreases in soil plasticity and elasticity (**Giovannini et al., 1988**). **Gabet (2014)** showed that soil burning detached erodible material in soil, resulting in an increase in wind-erodible sediment.

Increases in soil density following wildfire have been shown by numerous studies (e.g., **Giovannini et al., 1988**; **Andreu et al., 2001**; **Stoof et al., 2010**; **Jordán et al., 2011**; **Moody and Nyman, 2013**). In a typical example, measures of average bulk density of soils exposed in pits excavated following a prescribed fire by **Hubbert et al. (2002)** showed that burning resulted in 27% increases from 0.91 to 1.16 g/cm^3.

Another property of soils that changes in response to fire is the critical boundary shear stress or the stresses exerted by runoff that will result in the initiation of movement of a soil particle (**Moody et al., 2005**; **Moody and Martin, 2007**). The critical boundary shear stress,

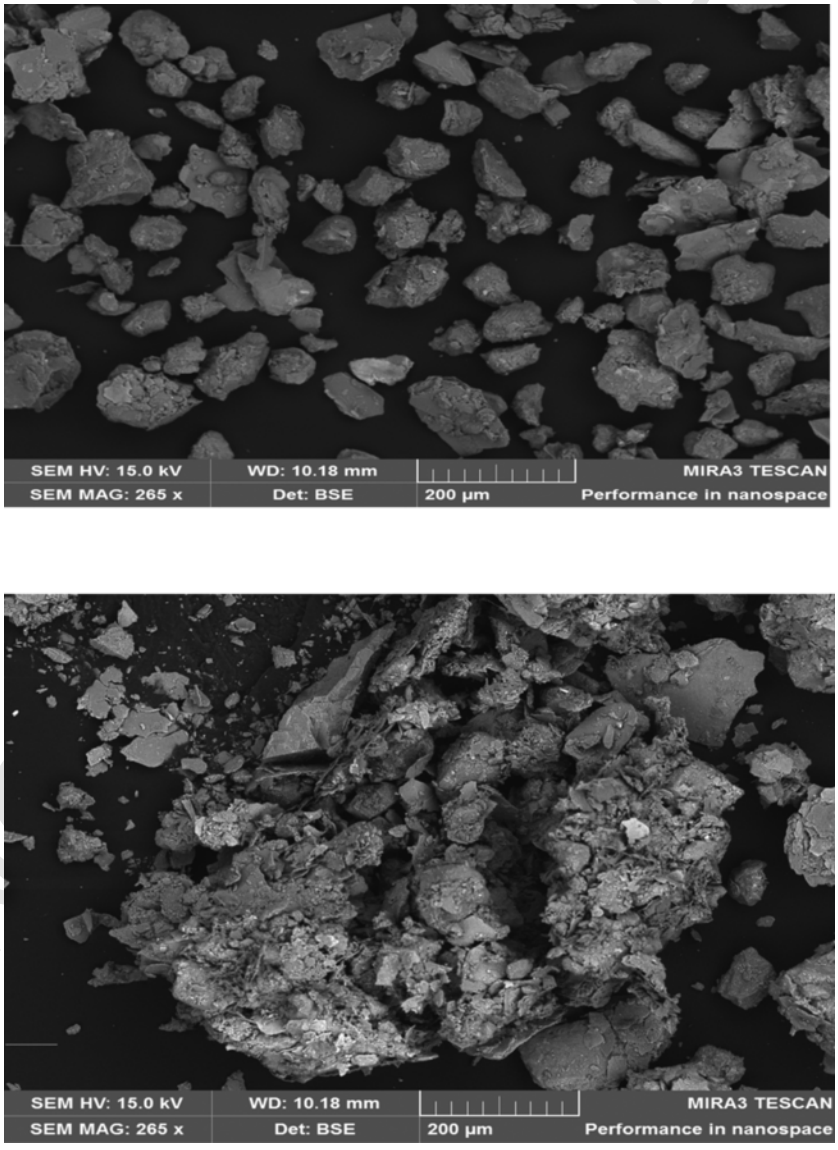


Fig. 10 Scanning Electron Microscope images of unburned (top) and burned (bottom) soils showing the resulting aggregates as a function of burning. Soils are derived from the Maroon Formation in Glenwood Springs, Colorado. Burning was done by open flame for 60 min using a propane torch with a 75 mm nozzle; the assumed effective temperature on soils > 500 °C.

UNCORRECTED PROOF

when linked with the associated critical shear velocity of water, provides a quantitative measure of soil erodibility. Critical shear velocities measured for unburned soils range between 0.022 and 0.19 m/s (**Elliot et al., 1989**; **Zhu et al., 2001**; **Moody et al., 2005**). **Moody et al. (2005)** found that at temperatures greater than 175 °C, critical shear velocity in burned soils varied with burning temperature and decreased with increases in temperature to an essentially constant range of 0.016–0.025 m/s. The shear stress reduction has been attributed to the deterioration of fine roots at the soil surface after a wildfire (**Nyman et al., 2013**).

Fire-induced rock weathering at macroscales is more readily observed than are the microscales of mineralogical and soil physical property changes. Spalling and cracking of rocks of all sizes, but especially boulders, are often observed in burned areas (**Kendrick et al., 2016**). Observations of spalling from boulders in burned areas in the mountains of southern Wyoming and the effect of fire upon building stones inspired Eliot **Blackwelder's, 1927** paper, "Fire as an Agent in Rock Weathering." Blackwelder postulated that fire's effect on rock is "…to expand the outer parts so suddenly that the tensile strength of the rock is exceeded radially and fractures are formed more or less parallel to the surface." **Fig. 11** is an example of spalling induced by wildfires that shows the plates and scales that Blackwelder described as characteristic of this type of weathering. **Blackwelder (1927)** postulated that this type of rock weathering could be a significant process in the wooded regions of the Rocky Mountains of the United States and Canada.

Birkeland (1984) also considered fire an important agent in the physical weathering of rock. He observed that stones exposed at the surface of an unvegetated rock glacier on Mt. Sopris in Colorado had oxidized surfaces, thick weathering rinds, and corners that were not well rounded despite their exposure to the elements for several thousand years. In contrast, stones in the adjacent, nearby forest lacked pronounced surface oxidation, had thin or no weathering rinds, and displayed well-rounded corners. That part of the forest had burned several decades earlier and presumably experienced other wildfires in both historic and prehistoric time. Noting that rock is a poor conductor of heat, **Birkeland (1984)** concluded that fire causes a thin heated outer layer that expands more rapidly than the rock at depth, producing ruptures in



Fig. 11 Fire-spalled granitic boulder within the 2008 Piute wildfire area in the southern Sierra Nevada south of Lake Isabella, California. The fire burned in a mixed conifer forest. The portable radio provides scale. Photograph by Jerome DeGraff, U.S. Department of Agriculture Forest Service.

UNCORRECTED PROOF

the form of thin sheets. In addition, **Clarke (1990)** conducted a series of field experiments involving ignition of natural fuels adjacent to or on granite boulders in the Pilbara region of Australia. He found that rock fracturing usually occurred between 4 and 8 min after the fire started and when temperatures exceeded 600 °C. More recently, **Shtober-Zisu et al. (2015)** found fire-induced spalling in 80–100% of carbonate rocks on Mt. Carmel, Israel.

Laboratory simulations have provided significant insight into the physical changes in rock induced by temperature. **Goudie et al. (1992)** and **Allison and Bristow (1999)** monitored changes in the modulus of elasticity of rocks with changes in temperature. The experimental work was performed on several different rock types including granite, marble, gabbro, and slate. As expected, heating resulted in different decreases in the elastic moduli of the samples, but **Goudie et al. (1992)** considered all decreases to be significant. Decreases in elastic moduli were measured in some rock types at temperatures as low as 200 °C. In addition, only one cycle of temperature change was sufficient to cause these decreases.

3 Process changes brought about by wildfire

3.1 Hillslope runoff and erosion

Runoff from burned areas can be either "saturation overland flow," where the soil is saturated from an impermeable surface up to the ground surface with continued application of rainfall, or "Hortonian" also known as infiltration-excess, overland flow, where rainfall input exceeds the infiltration capacity of the soil (**Horton, 1945**; **Dunne, 1978**; **Mirus and Loague, 2013**). Infiltration-excess overland flow is the most commonly measured of the two processes on burned hillslopes (**DeBano, 2000**, and references therein). Field monitoring by **Onda et al. (2008)** and **Schmidt et al. (2011)** support prior work indicating that infiltration-excess flow dominates most runoff response in burned areas. They recognize, however, that saturation overland flow also occurs on recently burned hillslopes, as was observed by (**Ebel et al., 2012**). **Schmidt et al. (2011)** found that although infiltration-excess overland flow is likely to dominate runoff events early in an individual storm, the temporal and spatial occurrences of both runoff processes will vary as a storm develops. **Ebel et al. (2012)** observed saturation overland flow at the ash-soil interface during a low-intensity rainstorm after a wildfire, but higher intensity storms produced infiltration-excess overland flow. As sites recover from wildfire, **Onda et al. (2008)** documented the occurrence of saturation overland flow later in the wet season as soil moisture increased. Thus, saturation overland flow likely is more common either during the latter stages of storms or later during the wet season when the moisture content of the soil profile increases.

Sealing of the mineral soil surface by post-fire vegetative ash can also reduce infiltration rates and promote runoff (**Campbell et al., 1977**; **Mallik et al., 1984**; **Durgin, 1985**; **Neary et al., 2005**; **Gabet and Sternberg, 2008**; **Onda et al., 2008**; **Larsen et al., 2009**). Sealing has been attributed to clogging of soil micropores by ash particles (**Mallik et al., 1984**; **Balfour, 2007**; **Woods and Balfour, 2008**), by swelling of ash particles within pores when wetted (**Etiegni and Campbell, 1991**), or by the formation of a surface crust by soil and ash compaction driven by raindrop impact (**Gabet and Sternberg, 2008**; **Onda et al., 2008**). However, in laboratory research, ash clogging in a pure sand matrix did not substantially influence water infiltration (**Stoof et al., 2016**).

In contrast to fire effects that promote hillslope erosion, a continuous layer of ash mantling hillslopes can absorb low intensity rainfall and thus inhibit runoff and erosion in the period shortly after a fire (**Cerdà, 1988**; **Martin and Moody, 2001**; **Balfour, 2007**; **Cerdà and Doerr, 2008**; **Woods and Balfour, 2008**) (**Fig. 7**). From rainfall simulation experiments, **Woods and Balfour (2008)** found that a 1–3.5 cm thick ash layer at a burned site in Montana provided additional storage capacity and infiltration, increased time to ponding and runoff, and reduced the final runoff rate. Similarly, in a series of field rainfall simulation experiments on ash overlying a sandy mineral soil at a burned site in Colorado, **Kinner and Moody (2008)** measured little or no discharge in response to rainfall intensities of less than about 20 mm/h and postulated that most of the rainfall was stored within the ash. However, at higher rainfall intensities, infiltration rates were limited by the presence of a coarser underlying mineral soil, rather than by the ash.

A spatially variable ash layer can also affect the development of flow paths and thus focus erosion. **Kinner and Moody (2008)** found that infiltration rates differed with the spatial variability of the ash; areas with thin or no ash cover became the primary flow paths. **Larsen et al. (2009)** documented increased runoff and sediment yields following removal of ash by simulated rainfall. **Woods and Balfour (2008)** found that the most significant post-fire runoff occurs after the ash layer has been removed or depleted, either during the first severe wind or rainstorm to impact an area or throughout subsequent rainy seasons.

Many researchers maintain that the passage of a fire results in a more friable, less cohesive, and more erodible soil (**DeBano et al., 1998**; **Scott et al., 1998**; Neary et al., 1999, 2005), with the changes to the soil depending on soil type and the temperatures reached during burning (**Guerrero et al., 2001**). Runoff-driven hillslope erosion rates can increase dramatically following wildfire (**Meyer et al., 2001**; Moody and Martin, 2001, 2009), with the increase generally attributed to the combined effects of loss of protective vegetative cover, decreases in water infiltration rates, and increases in the availability of readily eroded sediment (**Wondzell and King, 2003**). However, as easily mobile sediment is exhausted, coarser particles can be exhumed, increasing roughness and decreasing further erosion through hillslope armoring (**Rengers et al., 2016b**). Keep in mind, however, that in some cases, like the moist interior forest of western Canada, erosion rates do not significantly increase, but this lack of response is rarely reported.

Summarizing data from burned hillsides in the western United States, Australia, and South Africa, **Moody and Martin (2001)** found that erosion rates increased between 50 and 1000 times after fires, with a median increase of 160 times. For example, measured hillslope erosion rates increased between 20 and 200 times over pre-burn rates in Colorado (**Moody and Martin, 2001, 2002**; **Benavides-Solario and MacDonald, 2005**), 50–170 times in California (**Moody and Martin, 2001**), 6 times in the Coast Ranges of Oregon (**Roering and Gerber, 2005**), and up to 2 times in Valencia, Spain (**Campo et al., 2006**). From a compilation of 83 hillslope sediment yield rates measured within 2 years of fires in the western United States, **Moody and Martin (2009)** reported an average value of 8200 g/m^2/year, although amounts differed considerably depending on measurement method. Typical annual erosion rates reported for burned hillslopes are between 500 and 10,000 g/m^2/year, whereas typical rates for unburned, low severity burned, or recovered areas are between 1 and 50 g/m^2/year (**Benavides-**

UNCORRECTED PROOF

Solario and MacDonald, 2005; Campo et al., 2006; Shakesby and Doerr, 2006). Sediment delivery from a burn area in central Idaho was equivalent to several thousand years of the contribution from comparable unburned Idaho watersheds (Kirchner et al., 2001).

The highest sediment production rates are associated with high-severity burns and areas with high percentages of bare soil (Benavides-Solario and MacDonald, 2005). In the northern hemisphere, observations suggest erosion is greater on south-facing slopes than north-facing slopes because of reduced cohesion caused by long-term sun exposure (Moody and Martin, 2001; Ravi et al., 2006). In a comparison of the relative effects of loss of surface cover and soil-water repellency, Larsen et al. (2009) found that post-fire sediment yields were mostly determined by the loss of surface cover. Based on an analysis of data from throughout the western United States, Moody and Martin (2009) suggested that sediment availability may be more important than hillslope gradient or soil erodibility in predicting post-fire sediment yields.

There exists a distinction between diffusive water-driven erosion, which is dominated by overland flow or sheetwash, and incisive erosion, which is dominated by formation of rills and gullies, and channel incision. Ragan (2002) showed that areas characterized by diffusive erosion began to experience significant incisive erosion after wildfire. Loss of surface protection also results in gully and first-order drainage formation and expansion, as rills coalesce and concentrated runoff cuts deeply into unstable materials. Moody and Martin (2001) found that gully or drainage erosion following the 1996 Buffalo Creek fire in central Colorado accounted for 80% of the measured sediment movement, while rill and sheetwash erosion accounted for the remainder. More recent work by Ellett et al. (2019) found similar results. Hyde et al. (2007) found that gully formation tends to occur in basins that are extensively burned at high severities and have steep channel gradients. Incisive erosion is also prevalent in burned areas where debris flows are generated, as discussed in Section 3.3 below.

Rills are common erosional features formed after wildfire. They may form within minutes of even small rainstorms after wildfire (Wells, 1987; Schmidt et al., 2011). Post-storm observations indicate that rills approximately 5–20 cm wide formed primarily where loose, granular material, post-fire ash, and charcoal deposits remained, and extended to within 0.5–2.0 m of the drainage divide (Schmidt et al., 2011). Rills can be just a few centimeters deep, and in some cases, bases coincide with hydrophobic layers (Cannon et al., 2001a,b; Moody and Martin, 2001; Gabet, 2003; Sheridan et al., 2007). Atkinson (1984), however, described post-fire rills enlarging to gullies in southeastern Australia up to 60 cm deep. The erosion rate of rills in burned soils can be several orders of magnitude higher than on unburned soils, and the erosion rate appears to depend on runoff rate but has no correlation to the hillslope angle (Robichaud et al., 2010). However, the erosion rate of rills can be transient, with a higher initial sediment flux that decreases over time as sediment availability decreases (Robichaud et al., 2010).

Solifluction is a type of hillslope erosion that is observed after fires in regions underlain by permafrost. The post-fire loss of surface organic matter and its insulating effects contributes to thawing of permafrost, increasing the depth and duration of the freeze-thaw cycle, which in turn can lead to increased solifluction (Swanson, 1981). Working in the Yukon-Kuskokwim Delta in Alaska, Michaelides et al. (2019) used remote sensing of subsidence to conclude that wildfires had thinned the permafrost layer by a maximum of 5 m and that recovery to pre-fire thickness was expected to take 65–70 years.

3.2 Drainage network runoff and erosion

The loss of vegetation and decreased infiltration rates following wildfire can lead to marked increases in runoff from hillslopes during post-fire rain events, and this runoff inevitably focuses within the drainage network. Runoff rates measured 5 months after a burn near Valencia, Spain were three times the rates in similar unburned areas (Campo et al., 2006). Moody and Martin (2001) measured peak discharges 1 year after a fire in Colorado at levels 60 times those measured for a similar-sized rainstorm 3 years later after natural vegetative recovery.

Swanson (1981) hypothesized that base-flow stream runoff conditions are likely controlled by evapotranspiration and storage capacity of shallow soil, whereas peak flow conditions are controlled by soil infiltration rates. Following wildfire, both base flow (Kinoshita and Hogue, 2015) and peak flow (Neary and Gottfried, 2002) would be expected to increase, as evapotranspiration losses are lower, storage capacity of soils is diminished, and infiltration rates are lowered by the development of hydrophobic layers and accumulations of ash and fine materials in soil pores. The influence of evapotranspiration has been shown to increase discharge for several years post-fire, because the vegetation community recovering from wildfire has lower evapotranspiration than the established vegetation (Kinoshita and Hogue, 2011). Increased runoff is accompanied by increases in sediment transport rates, suspended sediment concentrations, dissolved sediment loads, and organic matter content, as dead vegetative matter and soil nutrients are flushed through the system (Walsh et al., 1995; Desilets et al., 2007; Malmon et al., 2007; Smith et al., 2011).

Erosive energy of flowing water within a channel may be approximated by stream power, Ω, defined as the rate of energy expenditure per unit length (W/m) (Legleiter et al., 2003):

$$\Omega = \rho g Q S \tag{1}$$

where ρ is the density of water (kg/m^3), g is the acceleration of gravity (9.81 m/s^2), Q is the discharge at the cross section being measured (m^3/s), and S is the channel gradient (unitless). Note that Eq. (1) is appropriate for the movement of clear water and does not include the effects of entrained sediment. Legleiter et al. (2003) measured stream power at 667 cross sections in the Yellowstone area following the 1988 fires and found that the minimum stream power increased with increasing percent burned area upstream of the cross sections (Fig. 12). They found that although calculations of minimum and average stream power increase with burned area, maximum stream power remains nearly constant.

Observations following wildfires yield different proportions of material eroded from hillslopes and channels. Moody and Martin (2009) found that approximately 75% of post-fire sediment yields come from channels, with the remaining 25% coming from the hillslopes, while Santi et al. (2008) found proportions closer to 90% channels and 10% hillslopes in recently burned debris-flow producing basins. Importantly, Staley et al. (2010) documented higher proportions of sediment moving from hillslopes into channels, rather than being flushed from channels, and attributed this to lower rainfall intensities than the erosion response documented by Santi et al. (2008) and Moody and Martin

UNCORRECTED PROOF

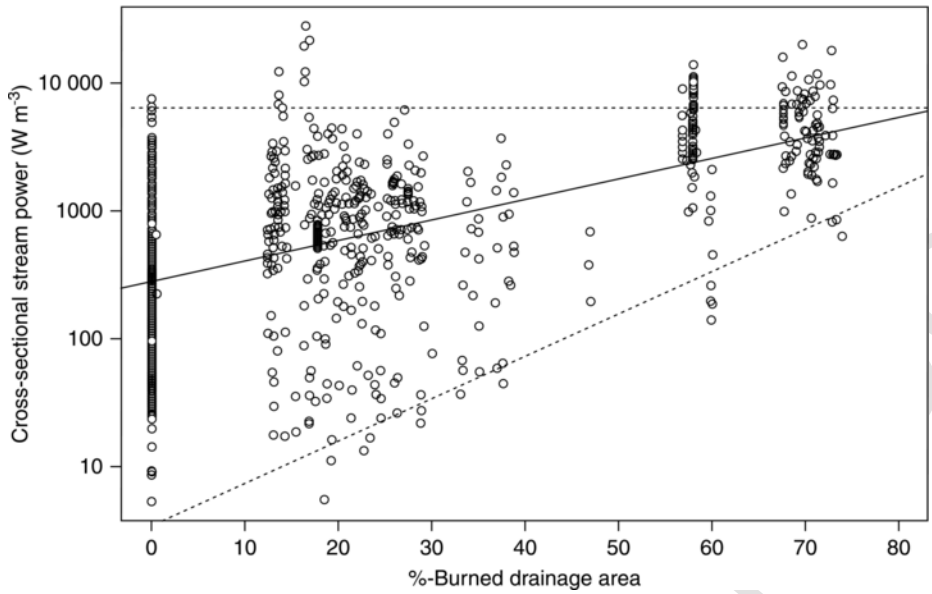


Fig. 12 Percentage area burned as a function of stream power. The solid line is a linear best-fit to the data, and dashed lines demarcate 5% and 95% confidence intervals. Minimum and average stream power increase with burned area, but maximum stream power remains nearly constant. Reproduced from Legleiter CJ, Lawrence RL, Fonstad MA, Marcus WA, and Aspinall R (2003) Fluvial response a decade after wildfire in the northern Yellowstone ecosystem: A spatially explicit analysis. *Geomorphology* 54: 119–136.

(2009). Moreover, as easily erodible material is moved by the initial rainstorms on a burned site, the sources of sediment can shift over time (**Tang et al., 2019a**).

3.3 Debris and mud flows

Debris and mudflows, which are rapidly moving masses of sediment, water, and air, can pose significant hazards to life and property. They can occur with little warning, exert great impulsive loads on objects in their paths, strip vegetation, block drainageways, damage structures, and endanger human life (**Iverson, 1997**). Following wildfire, enhanced erosive capacity is one of the most important drivers of debris flows and mudflows. Many studies (**Tognacca and Bezzola, 1997**; **Tognacca et al., 2000**; **Istanbulluoglu et al., 2003**; **Hungr et al., 2005**; **Hyde et al., 2007**; **Gabet and Bookter, 2008**) described possible theoretical frameworks that characterize the erosive processes that can result in channel erosion on steep hillslopes. However, the mechanics of the critical transition from sediment-laden water flow to debris flow is an active area of research (e.g. **McGuire et al., 2017**). Understanding the mechanics of this transition is important in burned areas, because debris flows are much more destructive than clear water flows (**Kean et al., 2016**).

Two primary processes have been identified for initiation of debris and mud flows from burned areas: erosion and entrainment of material by surface runoff (below); and mobilization of a discrete, infiltration-triggered landslide (discussed in **Section 3.4**) (**Coe et al., 2008**). The runoff-dominated process is more frequently reported in the literature; for example, 80% of 203 debris-flow producing basins included in a compilation of wildfire-related erosion events in the western United States showed this process (**Gartner et al., 2005**). The first rainstorm to impact a recently burned hillslope, when antecedent soil moisture would be minimal, may trigger debris flows (**Cannon et al., 2008**). Consequently, runoff-dominated processes are more prevalent in recently burned areas compared to longer-term infiltration processes that dominate unburned hillslopes. This is because, in contrast to unburned slopes where antecedent rainfall is commonly required to initiate shallow landslides and debris flows, in burned areas infiltration is diminished leading to more runoff.

In burned areas, many researchers have traced debris-flow deposits upslope through small gullies and into rill networks and noted the lack of a discreet landslide mass of a significant size at the heads of the flows (**Johnson, 1984**; **Parrett, 1987**; **Meyer and Wells, 1997**; Cannon et al., 2001b, 2003; **Jordan and Covert, 2009**). Rather, they found evidence of convergence and concentration of runoff within hollows and in low-order channels, as well as erosion and entrainment of significant amounts of surficial material. A common observation following fires and subsequent rainstorms is the runoff-induced development of rill networks on steeper hillslopes. **Scott (1971)**, **Parrett (1987)**, **Meyer and Wells (1997)**, and **Nyman et al. (2011)** emphasized that sediment input from hillslope rilling can be important in progressive sediment bulking. In some cases, the rills can be bordered by levees consisting of poorly sorted, unstratified, matrix-supported materials typical of debris flows (**Fig. 13**). **Wells (1987)** concluded that debris flows initiated from miniature (less than a few millimeters in thickness and a few square centimeters in area) soil slips at the heads of rills. The miniature soil slips occur in a saturated layer of soil a few millimeters thick and above a subsurface water-repellent zone (**Gabet, 2003**). Because these hillslope-generated debris flows (**Langhans et al., 2017**) can be diluted by additional water from overland and stream flow, they do not always persist or evolve into more destructive debris flows once they travel into channels (**Cannon et al., 2001a**).

UNCORRECTED PROOF



Fig. 13 Photograph of levee-lined rills produced by hillslope-generated debris flows as described by **Wells (1987)** and **Gabet (2003)**; 2019 Woodbury fire, Arizona. Photograph by Francis Rengers, U.S. Geological Survey.

UNCORRECTED PROOF

Extensive channel incision can lead to the formation of debris flows. For example, channel erosion from debris flows commonly extends to bedrock, with both runoff and sediment transported through the channel network. **Santi et al. (2008)** and **Jordan and Covert (2009)** found that most of the material in post-fire debris flows is eroded from channel beds within the channel network. Water-transported deposits were observed within low-order (0–2) channel reaches, and deposits characteristic of debris flows were observed farther down the drainage network. Based on these observations, **Meyer and Wells (1997)** and **Cannon et al. (2003)** concluded that surface runoff from a rainfall event erodes sediments from hillslopes and channels until sufficient material has been entrained, relative to runoff volume, for a debris flow to be generated. This process has been described as progressive entrainment (**Meyer and Wells, 1997**). An abundance of fine particles derived from loose, burned soil, dry-ravel deposits, and fine ash mantling the surface make burned areas particularly susceptible to this process (**Gabet and Sternberg, 2008**). Fine-particle entrainment may increase flow transport capacity and shear stress, enabling mobilization of coarser channel sediments, as reported by **Wan and Song (1987)** and **Gabet and Sternberg (2008)**.

Another mechanism for debris-flow initiation within channels (i.e. without either a landslide or a hillslope initiation source) is through en masse failure of sediment within channels (**Takahashi, 1981**). En masse failure can occur when sediment is preferentially deposited in low points in a channel and then fails through a retrogressive instability (**Kean et al., 2013**). This is a similar mechanism to ones suggested by **Wan and Song (1987)** and **Gabet and Bookter (2008)** that debris flows can be generated by runoff-initiated retrogressive erosion of gullies. En masse failure can be enhanced by stepped plunge pools. These are in-channel reaches that are susceptible to sediment storage and subsequent en masse evacuation (**Cannon et al., 2003**; **Rengers et al., 2018**) through headward migration of a knickpoint within an existing gully system (**Gabet and Bookter, 2008**). Another mechanism that would induce en masse failure of sediment is the “firehose” effect (**Johnson, 1984**), wherein runoff cascades over bedrock or cliffs and incorporates material from readily erodible downslope colluvium or channel beds (**Larsen, 2003**). Numerical modeling has been used to test progressive entrainment and en masse failure, and although both

mechanisms could potentially create debris flows within a channel network, en masse failure showed the surging behavior typically observed in real debris flows (**McGuire et al., 2017**).

Runoff-initiated debris flows from recently burned areas have been observed in a variety of environments, including the northern Rocky Mountains of the United States (**Klock and Helvey, 1976**; **Meyer and Wells, 1997**; **Parrett et al., 2003**) and British Columbia (**VanDine et al., 2005**; **Jordan and Covert, 2009**), the central and southern Rocky Mountains (Cannon et al., 2001a, 2001b; **Meyer et al., 2001**), the Uinta Mountains (**Larsen, 2003**) and Wasatch Front (**McDonald and Giraud, 2002**) of Utah, northeastern Oregon (William Russell, Oregon State University, written communication, 2003), the Huachuca Mountains of southeastern Arizona (**Wohl and Pearthree, 1991**), the Mazatzal Mountains of south-central Arizona (**Youberg, 2008**), southwest New Mexico (**Tillery and Rengers, 2019**), southern California (**Doehring, 1969**; Wells, 1981; **Wells, 1987**; **Morton, 1989**; **Spittler, 1995**; **Booker, 1998**; **Cannon, 2001**; Kean et al., 2011, 2019), and the Big Sur coast of California (**Cleveland, 1973**; **Johnson, 1984**). Debris flows generated through this process have also been described in the Swiss Alps (**Conedera et al., 2003**), France (**Fox, 2011**), and Australia (**Nyman et al., 2011**).

Although wildfires have burned in the rugged Coastal and Cascade Mountains of the Pacific Northwest, only a few post-wildfire debris flows generated through progressive bulking of runoff have been reported in this setting (**Wondzell and King, 2003**; **Wall et al., 2020**). **Swanson (1981)** and **Beschta (1990)** suggested that these areas are characterized by long-duration, low-intensity rainfall, thus infiltration rates are seldom exceeded, even after intense wildfires. **Wondzell and King (2003)** further proposed that rapid recovery of reduced infiltration rates caused by fires, high antecedent soil moisture, and rapid rates of vegetation regrowth after fires in these regions may explain the absence of overland flow. In addition, **Jordan and Covert (2009)** and **Jordan (2016)** found that although they do occur after exceptionally severe fire seasons, post-fire debris flows are rare in the mountains of British Colombia. Runoff in this environment is dominated by snowmelt, and runoff from even significant post-fire summer and fall rainstorms is unlikely to exceed typical snowmelt peaks.

Runoff-initiated debris flows have been produced by storms that occur up to 2–3 years after a fire, in response to both short-duration, high-intensity convective thunderstorms and longer-duration, lower-intensity frontal storms, and often during the first significant rainfall of the season (**Cannon et al., 2008**; **Jordan and Covert, 2009**; **Nyman et al., 2011**; **Santi and Morandi, 2013**). Approximately 2-year-recurrence-interval storms of either type may be sufficient to generate debris flows in recently burned basins. **Kean et al. (2011)** found that the occurrence of debris flows coincided strongly with short-duration (< 30 min) bursts of intense rainfall, while the time of peak stage for 79% of the debris-flow events was within just 10 min of peak 15-min rainfall intensity. Fire-related debris flows are generally triggered by rainfall durations at least an order of magnitude less than those described for unburned settings, and intensities are significantly lower (**Cannon et al., 2008**).

Most convective storms that generated runoff-initiated debris flows last less than about 3 h in duration, with most of the rain falling in less than 1 h at intensities between 1.0 and 32 mm/h (**Cannon et al., 2008**). Moreover, convective storms have triggered debris flows after as little as 6–10 min of rainfall with intensities of 50–75 mm/h in recently burned areas in south-central Colorado (**Cannon et al., 2008**) and after 7 min of rainfall with an intensity of 76 mm/h on the Big Sur coast of California (D. Reynolds, National Weather Service, written communication, April 8, 2009). Within longer duration storms that last for many hours, the rainfall intensity for short time periods (< 30 min) is typically the best predictor of a debris flow (**Kean et al., 2011**; **Staley et al., 2013**). In southern California, synoptic-scale weather patterns that cause post-wildfire debris flows have been associated with atmospheric rivers in many instances (**Oakley et al., 2017**). Even weak atmospheric rivers can lead to debris flows in southern California if they contain narrow cold frontal rain bands (**Oakley et al., 2018**).

Although differences in local rock, soil, and vegetation types, basin shapes and gradients, and burn-severity distributions will influence the debris flow rainfall intensity-duration thresholds (Cannon et al., 2008, 2010), only a few factors are needed to predict debris-flow likelihood throughout most of the western United States (**Staley et al., 2017**). Debris-flow likelihood can be successfully estimated using a logistic regression model (**Staley et al., 2013**) based on (1) slopes ≥ 23 degrees burned at moderate or high severity, (2) differenced normalized burn ratio, (3) rainfall accumulation, and (4) a soil erodibility parameter (the KF-factor) defined in the STATSGO database (Schwartz and Alexander, 1995).

3.4 Landslides

Shallow landslides have been documented in burned areas in a number of different settings, including southern California (**Scott, 1971**; **Morton, 1989**; **Cannon, 2001**), the Sierra Nevada of California (**DeGraff, 1997**), central Idaho (**Megahan et al., 1978**; **Meyer et al., 2001**; **Shaub, 2001**), north-central Washington (**Klock and Helvey, 1976**; **Helvey, 1980**), the Central Apennines in Italy (**Carabella et al., 2019**), western Greece (**Lainas et al., 2016**), and British Columbia (**Martin, 2007**). These landslides generally occur on soil- and colluvium-mantled hillslopes and involve thicknesses of sediment ranging from a few tens of centimeters to 6 m (**Fig. 14**). Reactivation of deep-seated, preexisting landslides has also been documented in burned watersheds (**Spittler and Wagner, 1998**), with landslides as large as over 1 km long and 1 km across.

In some cases, the failures mobilized into muddy slurries characteristic of debris flows. For example, debris-flow deposits that are thought to have originated from landslides related to forest-fire activity have been described in the Oregon Coast Range (**May and Gresswell, 2003**) and in east-central British Columbia (**Sanborn et al., 2006**). Of the 203 wildfire-related debris-flow producing basins surveyed by **Gartner et al. (2005)**, only 24 (12%) bore evidence of debris flows that originated exclusively from shallow landslides. Reports of exclusively landslide-triggered debris flows came from just four events: two in southern California following winter storms that impacted basins burned by fires the previous summer (**Morton, 1989**; **Cannon, 2001**), one in the Sierra Nevada of California 10 years after the fire (**DeGraff, 1997**), and one in response to a prolonged rainstorm with a greater than 100-year return interval, which culminated in a rain-on-snow event that triggered widespread flooding and landsliding throughout the northwestern United States (**Meyer et al., 2001**; **Shaub, 2001**). **Scott (1971)**, **Klock and Helvey (1976)**, **Cannon (2001)**, and Cannon et al. (2001b, 2003) presented evidence of debris flows triggered by both runoff and landslide processes in individual burned basins. These events account for 8% of the sample of 203 debris-flow producing basins of **Gartner et al. (2005)**. Importantly, **Cannon et al. (2001b)** found that when both processes resulted in debris flows, considerably more material was

UNCORRECTED PROOF



Fig. 14 Shallow landslide on hillslope burned the previous summer by the Missionary Ridge fire, outside Durango, Colorado. Photograph by Susan Cannon, U.S. Geological Survey.

UNCORRECTED PROOF

contributed to the flows from hillslope runoff and channel erosion than from the landslide scars. More recently, large landsliding that transitioned to a debris flow has been observed in a burn area of western North Carolina in the humid Appalachian Mountains (**Hill et al., 2020**).

It is necessary to consider possible changes to the balance of forces acting on hillslope soils when considering whether or not a wildfire is responsible for a landslide. Landslides occur when shear stresses acting on a slope equal or exceed the shear strength of the hillslope materials (**Scott, 1963**; **Sidle and Ochiai, 2006**). Shear stresses are imparted by gravity and the mass of the soil and structures and vegetation located thereon. Shear strength is the combined resistance to movement provided by friction on the shear surface (modified by pore-water pressure), and the cohesion of the mineral soil and plant roots. The balance between the driving and resisting forces depends primarily on slope angle, pore-water pressure, and the thickness and physical characteristics of the slope-mantling materials. The lack of vegetation after fire can result in more infiltration into the soil (after the effects of post-fire hydrophobicity have deteriorated) and less evapotranspiration. Therefore, more water contributes to increasing pore-water pressures during rainfall. In one study, the rainfall amount required to trigger landsliding was 20–30% lower on burned hillslopes than on unburned hillslopes (**Lainas et al., 2016**).

Three possible wildfire-related, landslide-triggering effects have been proposed, but few well-controlled data exist to evaluate them. First, increases in soil moisture after fires have been proposed as landslide triggers by **Klock and Helvey (1976)**, **Helvey (1980)**, **Megahan (1983)**, and Cannon et al. (1998), all of whom attributed the increase to a loss of vegetative interception and transpiration. Further evaluation is necessary, however, to determine if such increased moisture, and attendant pore pressures, is sufficient to trigger shallow landslides. Second, wildfire-induced tree mortality can lead to the decay of regolith-anchoring roots, and **Swanson (1981)** postulated that this decay could result in decreased soil cohesion and an increased probability of landsliding (see **Section 2.1**). A reduction in burned root tensile strength has been documented (**Jackson and Roering, 2009**; **Vergani et al., 2017**). Physical evidence that supports root decay-promoted failure is provided by **DeGraff (1997)**, who demonstrated that sheared roots exposed along the slip plane of a shallow landslide scar within an area burned by wildfire 10 years previously demonstrated failure of decaying, rather than live, material. The loss of root reinforcement after wildfire can create instability on steep slopes for decades (**Gehring et al., 2019**). And third, **Booker (1998)** and **Wondzell and King (2003)** suggested that increased peak flows after fire can contribute to accelerated bank erosion, leading to increases in bank-side failures.

Shallow surficial landslide failures in burned areas frequently occur in response to prolonged periods of storm rainfall, usually of a week or more duration (**Jordan and Covert, 2009**); periods of rapid snowmelt (**Schulz et al., 2006**; **Jordan and Covert, 2009**); or rain-on-snow events (**Scott, 1971**; **Meyer et al., 2001**; **Shaub, 2001**; **Cannon and Gartner, 2005**). Landslide-triggered debris flows that are activated during rain-on-snow events can be quite destructive, however, they occur infrequently.

Landslides have occurred during the first rainy season following a fire (**Morton, 1989**; **Cannon and Gartner, 2005**; **Jordan and Covert, 2009**), 1–2 years after fires (**Scott, 1971**; **Klock and Helvey, 1976**; **Meyer et al., 2001**), and up to 10 years (**DeGraff, 1997**; **Lainas et al., 2016**) and even decades (**Benda and Dunne, 1997**; **May and Gresswell, 2003**) after the fire. To establish if landslide failure can indeed be attributed to fire, separating the effects of the fire from those of the triggering meteorological conditions is needed.

At high latitudes, burning of taiga (boreal forests) can result in active-layer detachment slope failures because of an increase in active-layer thickness and pore-water pressures. These conditions result from reduced shading and albedo, with subsequent increase in absorption of solar radiation, compounded by reduced evapotranspiration and reduction in thickness of the insulating surface organic mat (**Lipovsky et al., 2006**).

3.5 Talus and dry ravel generation

Rock disintegration into soil-sized particles is temporarily increased due to the high temperatures produced by wildfire. Disintegration is often localized around cobbles, boulders, and outcrops. As described in **Blackwelder (1927)** and **Birkeland (1984)**, fire-caused spalling removes the outer millimeters to centimeters of the rock surface and reduces it to plates and wedges of rock. Spalled plates and wedges are readily weathered into smaller-sized particles, reducing the size of boulder fields and outcrops (**Swanson, 1981**). **Ollier and Ash (1983)** documented how trees burned near granodiorite boulders generated heat that caused spalls ranging in thickness from several millimeters to 15 cm.

Dragovich (1993) reported vertical fracturing associated with spalling in an arid region of Australia. He also concluded that spalling rounded the edges of larger boulders, and the surfaces of angular rock fragments, boulders, and outcrops exposed by spalling and cracking were subjected to further weathering from climatically controlled processes. **Selkirk and Adamson (1981)** examined sandstone affected by fire in the Blue Mountains of Australia and observed irregular spalling on the lower 3–4 m of a sandstone cliff, as well as small-scale pitting that made the rock more friable. They suggested that the pitting may allow water to more effectively erode the outcrop long after the fire. **Shtober-Zisu et al. (2015)** found that rock spalls serve as traps for sediment, ash, and charcoal for up to 20–30 years after the wildfire. **Dorn (2003)** surveyed rock surfaces before and after they were burned by a major wildfire in Arizona and demonstrated an increase in the rate of post-fire weathering and erosion of some rock types. This study is one of the few to quantify an increase in the amount of post-fire weathering and rock erosion.

Wildfire can also accelerate the rolling and bounding of rocks down steep burned hillslopes (**Swanson, 1981**; **DeGraff et al., 2015**). As a safety hazard, this process was tragically emphasized in 1999 when a firefighter was killed after being struck by a bounding rock (**Fahy and LeBlanc, 2000**). During the 2008 Chalk fire in Los Padres National Forest in California, US Highway 1 and other roads were closed because of this safety concern. The Incident Command Team recognized that the slopes above them continue to have rolling rocks and debris. Highways in British Columbia are frequently closed due to rockfall during fires, such as the 2007 Springer fire near Slocan, the 2009 Tyaughton fire near Lillooet, and many others (P. Jordan, oral communication, 2011). Movement of rocks occurs when fires consume woody stems that would normally catch and hold rocks on unburned hillslopes or when fire incinerates the organic material, including fine root mats that bind the soil and rock (**Swanson, 1981**).

Many of the same factors that contribute to downslope rolling and bounding of rocks on burned hillslopes also contribute to the dry ravel process. Dry ravel is an important post-fire process that can be observed both during and after the passage of a fire (**Swanson, 1981**; **Lamb et al., 2011**; **Florsheim et al., 2016**). It is the downhill movement of individual regolith and organic particles under the influence of gravity and without the effect of water (**Swanson, 1981**). This process occurs in response to drying of the soil, combustion of soil-binding organisms, bioturbation, and removal of stems that buttress material on hillslopes prior to the fire. Strong winds can move ash and fine-grained material during or after the fire (**Swanson, 1981**; Cannon et al., 2001a, 2001b; **Van Wagtendonk, 2006**; **Schmidt et al., 2007**). **Cannon et al. (2001b)** reported that in the days following the 2001 South Canyon fire in Colorado, clouds of white dust towered above the burned area. The ash and loose, friable, and exposed burned mineral soil on the hillsides were transported by wind and dry ravel and redeposited on hillslopes and in the tributary drainages. This process results in aprons of material at the angle of repose mantling and smoothing hillslopes, lining channel banks, and filling in small channels (**Fig. 4**). The readily erodible dry-ravel deposits are frequently entrained and transported by rainfall-triggered runoff. Although thick accumulations of dry ravel in channels are thought to contribute to the generation of debris flows following fires (**Wells, 1987**; **Florsheim et al., 1991**; **Schmidt et al., 2011**), documentation of debris flows in settings without dry ravel led **Cannon and Gartner (2005)** to point out that it is not a prerequisite for their generation.

3.6 Recovery and return to original conditions

In the years following a wildfire, some conditions and processes can gradually return to their pre-fire states, as vegetation is reestablished, water-repellant layers dissipated, a surface lag of coarse sediment that protects underlying materials develops, and sediment generated during and after the burn is transported through the system (**Martin and Moody, 2001**; **Legleiter et al., 2003**). The time frame during which the landscape is affected by a fire has been termed the "window of disturbance" (**Prosser and Williams, 1998**) (**Fig. 15**). This interval is different for each process and for different climates and geomorphic settings.

Benavides-Solario and MacDonald (2005) measured vegetation recovery in the Colorado Front Range by plotting the percent of bare soil remaining after a fire (**Fig. 16**). They found that recovery was rapid at first, with 50% vegetative cover established within 2 years, regardless of the burn severity. The process slowed over the next few years, with moderate and high severity burns recovering less (80% vegetative cover after 7 years) than low severity burns (over 90% vegetative cover after 7 years). Vegetative recovery in humid climates is faster than in arid or semiarid climates (Moody and Martin, 2002). Recovery may be on the order of 50–500 years in more extreme climates such as alpine terrain (**Sass et al., 2012**).

Windows of disturbance have been estimated or measured by a number of researchers (**Table 2**). Although a range of responses has been shown, sediment discharge generally seems to return to pre-burn levels after approximately 3–4 years. Exceptions to this time frame may occur in colder regions, where fire affects permafrost and active layer depths. Recovery related to slope stability in burned taiga zones has been estimated at 25–50 years, because of strong dependence on reestablishment of spruce and feathermoss communities (**Vierick, 1982**; **Lipovsky et al., 2006**).

4 Landform changes brought about by wildfire

4.1 Hillslope lowering

Measurements of hillslope and channel erosion rates following wildfires can be extrapolated over the landscape to calculate a rate of hillslope lowering over time, and such measurements indicate that wildfires are an important agent of landscape changes, when linked with sufficient rainfall. For example, **Jackson and Roering (2009)** reported that the amount of post-fire dry-ravel accumulations in the first few months of a low-order valley in the Oregon Coast Range was equivalent to a slope-averaged landscape lowering of 2.5 mm. **Abrahams et al. (2018)** used radionuclides to measure post-wildfire erosion in the Front Range of Colorado, measuring a mean sediment loss on north-facing slopes of 2 mm over a 1–3 year period, a rate that is 1–2 orders of magnitude higher than previously reported in this region. **Moody and Martin (2009)** found some of the highest post-fire sedimentation rates comparable to long-term sediment yields from major watersheds of the world, with periods of increased erosion following wildfires being responsible for between 25% and 50% of the overall basin sediment production,

UNCORRECTED PROOF

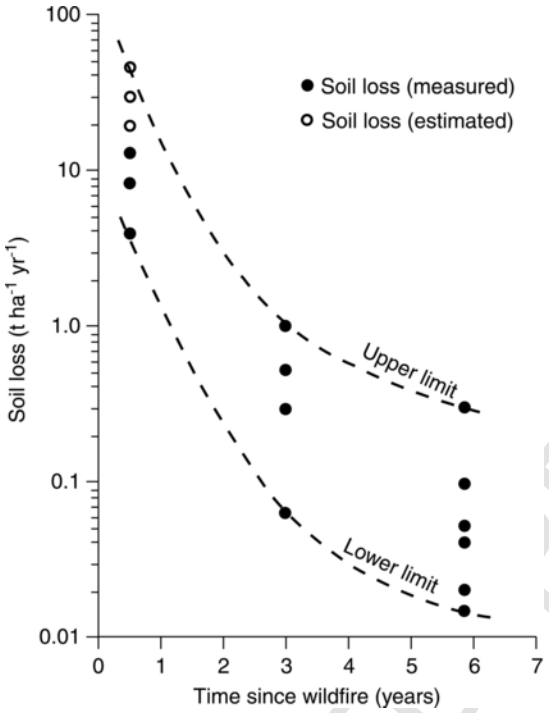


Fig. 15 Plot of measured soil loss as a function of time since wildfire. Soil loss rates are several orders of magnitude higher immediately post-fire but return to pre-fire levels within several years. Reproduced from Shakesby RA, and Doerr SH (2006) Wildfire as a hydrological and geomorphological agent. *Earth-Science Reviews* 74: 269–307.

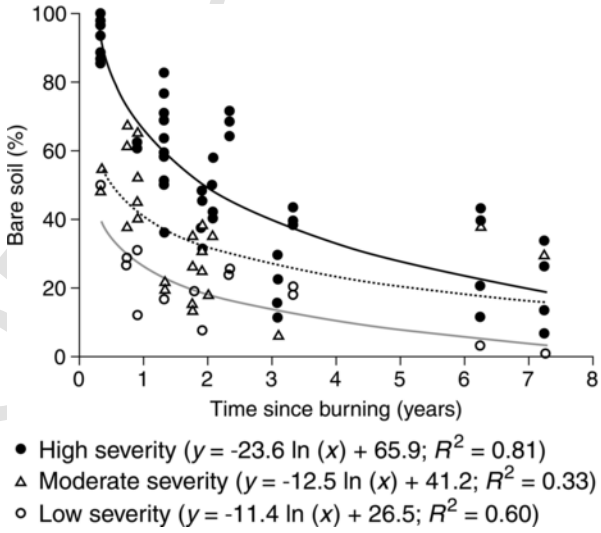


Fig. 16 Percentage of bare soil following a burn, as related to burn severity. Recovery to pre-burn conditions (80–90% vegetated) takes approximately 7 years. Reproduced from Benavides-Solario J, and MacDonald LH (2005) Measurement and prediction of post-fire erosion at the Hillslope scale, Colorado Front Range. *International Journal of Wildland Fire* 14: 1–18.

depending on the setting. **Lavé and Burbank (2004)** documented that an increase in the occurrence of fires during the last century in the Transverse Ranges of southern California has augmented sediment delivery to debris basins by an average of > 60%, and to individual basins by up to 400%. They also found that hillslope-produced sediment may be stored in second-order and higher channels until major storms mobilize it, such that the larger fluvial network dampens the episodic, fire-induced hillslope sediment pulses produced within small watersheds.

UNCORRECTED PROOF

Table 2 Examples of measured recovery periods ("windows of disturbance") reported in the literature.

Window of disturbance	*Location*	*Conditions measured or observed*	*Reference*
2–4 years	Rocky Mountains, United States	Sediment yields	**Martin and Moody (2001)**; **Legleiter et al. (2003)**
2–3 years	Western United States and southern California	Runoff-dominated debris flows in severely burned basins in	**Cannon and Gartner (2005)**
1–3 years	Western United States, British Columbia, Italian Alps	Debris-flow volumes	**Santi and Morandi (2013)**
3–4 years	Oregon, United States	Erosion rates	**Robichaud and Brown (1999)**
3 years 7 years 14 years	Southwest United States	Low, moderate, and high severity burn sediment yields	**DeBano et al. (1998)**
3 years	Northern Portugal	High severity burn, runoff response in maritime pine forest	**Walsh et al. (1995)**
6 years	Oregon, United States	High severity burn, reduced infiltration rates in dry pine forests	**Dyrness (1976)**
1–3 years Up to 30 years 30 + years	Colorado, United States	Sediment flux Water discharge Regrowth of trees	**Moody and Martin (2001)**

4.2 Channel aggradation

Increased erosion and transport of material from hillslopes following wildfires may result in immediate channel aggradation, particularly in lower-order channels in the higher reaches of drainage basins. For example, it has been observed that in tectonically active mountains with gradients exceeding the angle of repose, colluvium can move into channels via dry ravel as loose sediment stored behind vegetation is released once the vegetation burns (**Lamb et al., 2011**; **DiBiase and Lamb, 2013**). After the first rain events, deposits observed in low-order channels can consist of both debris-flow and alluvial-transported materials. These channels are typically transport limited, in that sediment supply exceeds the stream carrying capacity (**Santi and MacAulay, in review**), and the alluvial deposits are coarse bedload material. Suspended sediments can be transported much farther, and streams draining burned areas can remain muddy for several years after a fire. Sediment moving in smaller channels can deposit as fans in the higher order stream channel valleys at tributary junctions, creating upstream low-gradient reaches and downstream high-gradient reaches that provide useful habitat diversity for aquatic ecosystems (**Benda et al., 2003**).

Over time, streamflow may rework and remove both the debris-flow and bedload channel deposits. Sediment has been observed to move in stages, with smaller particles and ash moving the farthest after rainstorms immediately after fire, while coarse sediment is stored in fluvial channels (**Reneau et al., 2007**). In subsequent years, the coarser sediment continues to move downstream during storms (**Reneau et al., 2007**), and this sediment may be redeposited farther down the drainage network where the stream gradient drops, creating stream terraces (**Meyer and Wells, 1997**). **May (2002)** documented the reworking of post-fire debris-flow deposits by subsequent streamflow; sediments are carried downstream, resulting in channel aggradation and formation of a stream bed of coarse gravel. **Meyer and Wells (1997)** postulated that surface soils are quickly stripped of the silt and clay necessary for debris-flow generation, such that streamflow processes dominate the response to later rainstorms.

Keller et al. (1997) found that most of the mobile sediment from the burn areas was transported and deposited in stream channels below the burned area within 2 years after fire. Similarly, **Lane et al. (2006)** saw a tenfold increase in sediment transport in the first year following fire and 2–4 times increase in the second year, due to increased stream flows after wildfire. **Moody and Martin (2001)**, however, documented that two-thirds of mobilized sediment after a fire in the Colorado Front Range was still stored within a few miles of the burned watershed after 4 years, and they estimated a residence time for the sediment in that location of over 300 years. Wherever material is deposited following a fire, the channel slope can decrease (**Murphy et al., 2019**), and this reduces the channel capacity, creating the potential for lateral migration and overbank sedimentation, such as documented in Yellowstone National Park by **Meyer et al. (1995)**.

4.3 Channel incision

The pulse of runoff generated in recently burned areas, enhanced by root deterioration, can result in significant channel incision. Post-fire incision can result in newly formed or rejuvenated gullies after wildfire in unchannelized colluvial hollows (**Eccleston, 2008**; **Gabet and Bookter, 2008**; Hyde et al., 2007, 2014), and the development of channel steps or knickpoints (**Cannon et al., 2003**; **Rengers et al., 2018**). **Legleiter et al. (2003)** showed that elevated post-fire stream discharge, measured as stream power (Eq. **1**), when combined with decreased sediment supply once vegetation begins to restabilize watershed slopes, induced a period of channel incision. Measured channel incision in the years following wildfire ranges up to 4 m in Nevada (**Germanoski, 2002**) and up to 6 m in Montana (**Parrett et al., 2003**). **Legleiter et al.**

UNCORRECTED PROOF

(2003) also concluded that channel incision was the dominant response in second- and third-order channels following the Yellowstone National Park fires, and this response stemmed directly from increased stream power. Eaton et al. (2010), however, found that stream channel changes after a severe 2003 fire in British Columbia were due to root deterioration on channel margins following burning of riparian vegetation, not due to increased peak flow. The duration of flow in a channel is also important for the overall channel incision. For example, channels in burned watershed in Colorado experienced greater incision and channel widening after a long flood with a lower peak discharge than a short flood with more than double the peak discharge (Brogan et al., 2017). Meyer et al. (1995) found that over a period of thousands of years, warmer drier climate intervals in Yellowstone were associated with increased fire occurrence, which led to meander-belt narrowing and channel incision during infrequent large floods. Post-fire incision can have long-lasting influences on stream ecology; for example, channel erosion can disrupt or eliminate macroinvertebrate communities for up to a decade (Leonard et al., 2017).

4.4 Alluvial and debris fans

Increases in runoff and erosional processes within burned watersheds immediately after fire, and persisting for several years, can result in increased sedimentation on tributary alluvial and debris fans (Benda, 1990; Benda and Dunne, 1997; Jordan, 2016; Kean et al., 2019). Meyer et al. (2006) demonstrated from alluvial fan stratigraphy and dating of ash and charcoal deposits that, between 4000 and 5000 years ago, approximately 30% of alluvium in Yellowstone subalpine settings consisted of fire-related deposits and that approximately 50% of sediment in fans in central Idaho could be attributed to fire. Tributary alluvial fans and mainstem floodplains aggraded between 2000 and 500 years ago in response to fire- and storm-induced erosion in the high-elevation conifer forests of northern Yellowstone National Park (Meyer et al., 2008). In central Idaho, severe fires that burned between 950 and 1150 CE resulted in large debris flows (Meyer et al., 2006). In this area, approximately 25% of total fan aggradation over the past 2000 years has been attributed to these debris flows (Meyer et al., 2008). Rapid, post-fire aggradation of fans and valley floors within the Sacramento Mountains of southern New Mexico and the San Juan Mountains of southwestern Colorado have also been documented during this period (Frechette and Meyer, 2009; Bigio et al., 2010). Post-fire fan deposition has been seen in other warm portions of geologic record, such as the Cretaceous (Muir et al., 2015).

Sediment is generally delivered to fans episodically following fires depending on storm rainfall conditions, and periods of increased deposition have also, in some cases, been found to be related to prevailing climatic conditions. The depositional fans preserving post-fire sediments can take the form of either sheetflow or thicker debris flow deposits (Riley et al., 2015). Meyer et al. (2008) documented fire-related debris flows associated with the ca. 1150 CE mega-drought in Yellowstone. A major pulse of debris-flow activity has been documented during a drought during the late 1200s AD in the Sacramento Mountains of New Mexico (Frechette and Meyer, 2009).

4.5 Talus cones and rock movement

Wildfires can accelerate the development of talus cones. Large numbers of rocks roll down steep burned hillslopes and come to rest in low-gradient areas, such as valley floors or roadbeds. The rocks will either increase the volume of existing talus deposits or initiate new talus cone formation. Talus cone development was rapid enough during the July 2008 Ukonom Complex wildfire in northern California that a dust cloud was generated (Fig. 17). This example is one of a number of talus cones that formed in this fire, necessitating road closure to all traffic except emergency vehicles (Ivan Erskine, oral communication, 2009).

Rock weathering and erosion by wildfire produce other distinctive landforms. Working on slopes in arid and semiarid environments of the Pilbara in Western Australia, Dragovich (1993) found that the vegetation-controlled distribution of fire concentrated the physical breakdown of rock in particular parts of the landscape. These locations include the base of boulder slopes, around vegetated sediment areas at the crest of some slopes, and on patches within some boulder slopes themselves. Consequently, the presence of boulders and rock fragments and the resulting susceptibility of the slope to other erosional agents produced a distinctive landscape pattern. Dorn (2003) concluded from his observations of boulders in a fire-affected area in Arizona that single hot fires could reduce boulders to a number of small clasts. If this occurs, the resulting eroded products would be more quickly moved by gravity- and fluvial-driven processes, expanding the influence of wildfire on landscape evolution. While these examples are revealing, Shakesby and Doerr (2006) point out that the number of studies and their geographic coverage to date are limited.

5 Applications of geomorphology in burned areas

5.1 Post-wildfire team studies

Post-wildfire changes in geomorphic processes have long-term implications for landscape development and more immediate societal implications. Flooding, sedimentation, debris flows, and rockfall may threaten the lives, homes, and economic well-being of people near a burned area. Most of these threats are acute during the first rainy season following the wildfire (Cannon and Gartner, 2005; Thode et al., 2006). In the western United States, there is a nexus of public land subject to wildfires and increasing population (DeGraff et al., 2007). Consequently, land management agencies of the US government, such as the Forest Service, National Park Service, and Bureau of Land Management, require assessment of burned areas to determine whether emergency situations affecting resources or human health and safety exist. The assessment is carried out by teams consisting primarily of geologists, hydrologists, soil scientists, and other specialists. Within the federal jurisdictions, depending on the agency involved, they are designated Burned Area Emergency Response (BAER) teams or Emergency Rehabilitation and Stabilization (ERS) teams (Thode et al., 2006). Some state agencies have also developed similar assessments for state lands, such as the Watershed Emergency Response Team (WERT) in California.

BAER teams assess the potential changes in geomorphic processes caused by wildfire on vegetation and soil and provide recommendations for mitigation. They prepare maps showing relative burn severity as differences between the pre-fire vegetation and what

UNCORRECTED PROOF



Fig. 17 Rocks forming a talus cone on a road in northern California during the Ukonom Complex wildfire near Orleans, California. Rock deposition was sufficiently active that dust rose from the developing talus cone. Photo by Ivan Erskine, USDA Forest Service.

remains after the fire. Initial burn severity maps are commonly produced on the basis of reflectance values detected by space-based sensors. These burn severity maps (**Fig. 18**) guide field investigations that refine the initial map into the final burn severity map (**Parson et al., 2010**). Specialists use this mapping to model potential changes and identify possible onsite and downstream impacts. Any change in runoff, debris-flow potential, or other geomorphic process is linked to any highways, pipelines, homes, or other infrastructure that might be adversely affected. This assessment must be rapidly conducted to ensure that any countermeasures can be implemented before the first damaging storms (**DeGraff et al., 2007**). Countermeasures may include emergency evacuation plans for people threatened by flooding, closure of trails or roads where rockfall or debris-flow hazards may increase, or construction of debris-flow deflection walls. **DeGraff (1994)** provides an illustration of BAER assessment and implementation of countermeasures following a 1990 wildfire near Yosemite National Park.

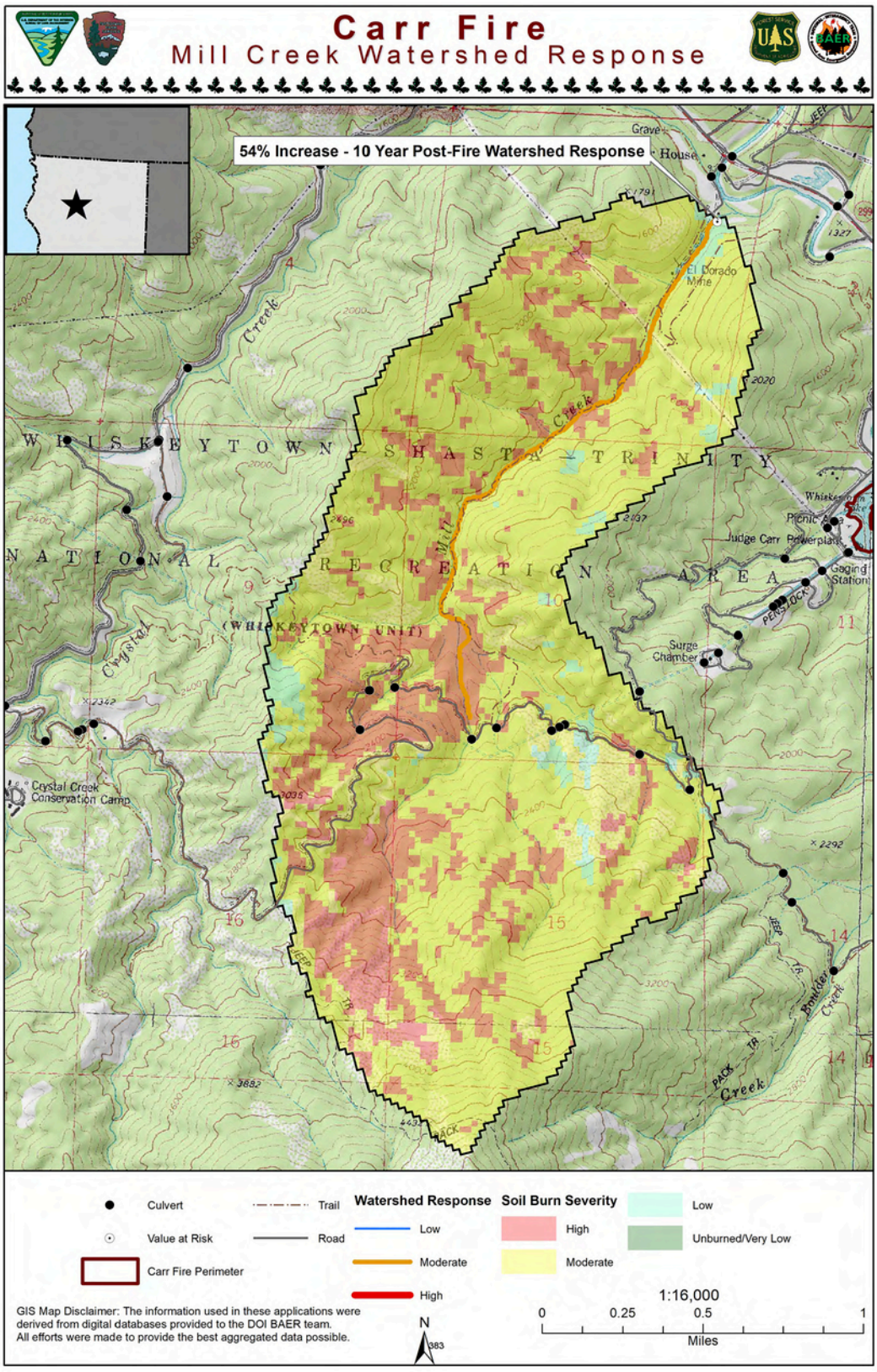
5.2 Debris-flow hazard assessments

Debris flows are among the most hazardous consequences of rainfall on recently burned hillslopes. Within the burned area and downstream, the powerful force of rushing water, soil, and rock can destroy buildings, roadways, culverts, and bridges, and can cause injury or death (**Kean et al., 2019**). Debris flows in recently burned areas are particularly dangerous because they can be generated in response to very little rainfall and in places where flooding or debris flows have not been observed in the past (**Cannon et al., 2008**). For example, a storm on Christmas Day 2003 triggered debris flows from many of the steep basins burned the previous fall in the San Bernardino Mountains of southern California (**Brock et al., 2007**). Sixteen people were killed by this event, and billions of dollars were spent on cleanup and repairs (**Chong et al., 2004**). A similarly devastating event occurred on January 9, 2018, in Montecito, California, where a narrow cold frontal rain band moved across the Thomas fire, which was still burning at the time of the rainfall, resulting in widespread debris flows and 23 fatalities (**Kean et al., 2019**).

The past few decades of work on post-fire debris flows have often pointed to initiation mechanisms that are consistent with debris flows initialized from overland flow runoff, rather than a discrete landslide source (**Cannon, 2001**; **Cannon et al., 2008**; **Santi et al., 2008**; **Kean et al., 2011**; **Staley et al., 2014**). These debris flows typically initiate from rainstorms in the first rainy season or two following a fire. Modeling studies suggest that overland flow runoff can generate debris flows within a channel network through either progressive entrainment of sediment or en masse failure (**McGuire et al., 2017**). Moreover, a critical amount of flow (characterized as a dimensionless unit discharge) can be used to estimate where and when debris flows will develop during rainstorms (**Rengers et al., 2019**; **Tang et al., 2019b**; **Tillery and Rengers, 2019**). Under these conditions, traditional slope stability analyses would not appropriately characterize debris-flow hazards. If the concern is the response of a landscape over longer time frames, for example, long enough to account for the effects of root decay, the impacts of failure of infiltration-triggered landslides will need to be evaluated (see **Section 3.4**).

In recent years, tools have been developed to provide land managers and emergency responders with science-based information to identify areas with the greatest danger to human life and infrastructure after fires and to effectively design and implement mitigation measures and emergency response plans (**Prochaska et al., 2008b**). For example, the US Geological Survey uses empirically based logistic regression models to estimate debris-flow likelihood (**Staley et al., 2017**) and debris-flow volumes (**Gartner et al., 2014**) in burned basins throughout the western United States. These models can be readily implemented in a GIS to produce maps showing potential debris-flow hazards for given storm events (**Fig. 19; U.S. Geological Survey, 2020**). In addition, physically based models have been developed to successfully estimate the timing of debris-flow peaks in burned watersheds using fully distributed hydrologic models (Rengers et al., 2016a, 2019).

UNCORRECTED PROOF


Carr Fire
Mill Creek Watershed Response
54% Increase - 10 Year Post-Fire Watershed Response
Culvert
Value at Risk
Carr Fire Perimeter
Trail
Road
Watershed Response
Low
Moderate
High
Soil Burn Severity
Low
High
Moderate
Unburned/Very Low
GIS Map Disclaimer: The information used in these applications were derived from digital databases provided to the DOI BAER team. All efforts were made to provide the best aggregated data possible.
N
1:16,000
0 0.25 0.5 1
Miles

Moreover, debris-flow surge behavior can be simulated, when hydrologic models are paired with sediment transport equations (McGuire et al., 2016, 2017). Field reconnaissance following debris flows has also led to information that can be used for quantitative assessments of infrastructure risk (**Kean et al., 2019**). Work has even been done to estimate debris-flow probabilities in watersheds prior to a wildfire (**Staley et al., 2018**). Similarly, the return period of debris flows has been estimated based on probability of triggering rainfall events and debris-flow mobility (**VanDine et al., 2005**).

Several studies have explored approaches to model debris-flow runout behavior. Cannon et al. (2009) developed a method to estimate the area that may be inundated by the volumes of material estimated for each basin, using an approach modified from that developed by **Iverson et al. (1998)** to map inundation areas of lahars (debris flows generated from volcanoes). This involved a terrain analysis to identify the position within drainage networks where the onset of continuous deposition is likely to occur as a function of channel gradient and degree of confinement, combined with a set of regression equations that relate debris-flow volume to cross-sectional and planimetric inundation areas (**Bernard, 2007**), to map potential inundation areas. The method identifies the area that will be covered or filled with debris-flow deposits and is appropriate for single debris flows that travel from steep, tightly confined drainage basins into wide channels on relatively low gradient surfaces. As a variation on the approach described above, **Elliott et al. (2004)** coupled the models for post-fire debris-flow susceptibility (**Cannon et al., 2010**) with a two-dimensional algorithm for simulating the depth and travel of mud flows (**O'Brien, 1993**) to characterize debris-flow hazards from recently burned basins in Colorado. Finally, **Prochaska et al. (2008a)** developed a method for estimating how far debris flows might travel on alluvial fans. The method predicts debris-flow runout as an angle of reach from the mid-point elevation of the drainage channel, measured from the apex of the alluvial fan to the top of the drainage basin, to the end of the runout zone.

6 Summary

Wildfire is a widespread phenomenon that is expected to increase in extent and severity in the future, due to fuel accumulations, shifting land-management practices, and climate change. Primary effects of fire include removal of soil-mantling vegetation and litter, deposition of ash, creation or destruction of water-repellent soils, and physical weathering of boulders and rock.

Wildfires can be an important agent of landscape change when linked with sufficient rainfall. Decreased soil permeability following wildfire, coupled with removal of vegetation and litter, can result in lower interception and infiltration rates, and increased runoff. These changes may be accompanied by increased erosion through sheetwash, rilling, and dry ravel of granular soil and rock material, and increasing intensity and frequency of floods, debris flows, rockfall, and landslide movement. Studies have shown that the window of disturbance to the landscape caused by wildfire is typically on the order of 3–4 years, although some effects in some environments can persist up to 30 years or even longer in rare instances. Moreover, sedimentological records show that the incidence of wildfire increases during warmer climate periods.

Refining and expanding our understanding of fire as an agent of landscape change requires a consideration of current and past variability in the occurrence of fires. This variability may involve more than consideration of fire return periods for the vegetative communities present in a particular landscape. At the landscape scale, the temporal and areal occurrence of wildfires is not solely controlled by the character of the vegetative community being burned. It is also influenced by climatic shifts and anthropogenic changes. Consequently, understanding the influence of fire on the landscape will require careful evaluation of past and present wildfire regimes.

Dedication

We dedicate this article to our wonderful colleague, Jerry DeGraff. Jerry was a coauthor of the first edition of this work, and while he intended to be coauthor for this edition, he unfortunately passed away before he could begin his contribution. Having spent a long and productive career with the US Forest Service, Jerry was deeply involved in post-wildfire monitoring and recovery efforts, and his publications are a valuable and insightful legacy.

Fig. 18 Example burn severity figure created by the Burned Area Emergency Response (BAER) team assessing the 2017 Goat and Sliderock Fires in Montana. Areas of high burn severity with the greatest loss of vegetation and impact to soil conditions are shown in red. Image prepared by the U.S. Department of Agriculture Forest Service.

UNCORRECTED PROOF

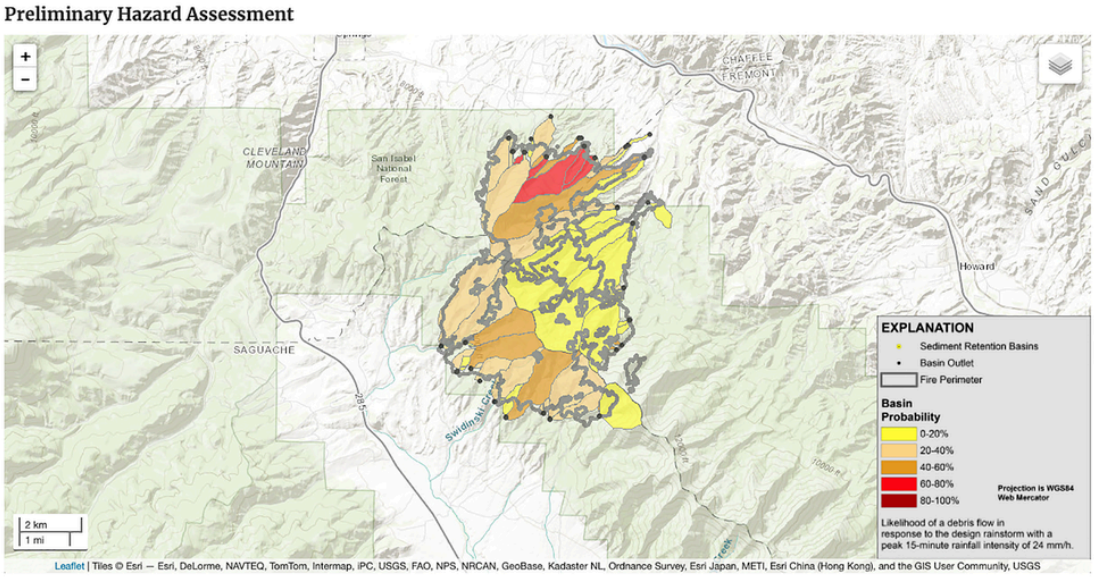


Fig. 19 Example of a Debris Flow Hazard Assessment for the Grizzly Creek fire in western Colorado. U.S. Geological Survey (2020) Grizzly Creek Fire Debris-flow Assessment. Retrieved from https://landslides.usgs.gov/hazards/postfire_debrisflow/detail.php?objectid=317.

References

Abe, K., Ziemer, R.R., 1991. Effects of Tree Roots on Shallow-Seated Landslides. USDA Forest Service, Pacific Southwest Forest and Range Experiment Station, 11–20, General Technical Report PSW-GTR-130.

Abrahams, E.R., Kaste, J.M., Ouimet, W., Dethier, D.P., 2018. Asymmetric hillslope erosion following wildfire in Fourmile Canyon, Colorado. Earth Surface Processes and Landforms 43, 2009–2021.

Allison, R.J., Bristow, G.E., 1999. The effects of fire on rock weathering: Some further considerations of laboratory experimental simulation. Earth Surface Processes and Landforms 24, 707–713.

Andreu, V., Imeson, A.C., Rubio, J.L., 2001. Temporal changes in soil aggregates and water erosion after a wildfire in a Mediterranean pine forest. Catena 44, 69–84.

Arocena, J., Opio, C., 2003. Prescribed fire-induced changes in properties of sub-boreal forest soils. Geoderma 113, 1–16.

Atchley, A.L., Kinoshita, A.M., Lopez, S.R., Trader, L., Middleton, R., 2018. Simulating surface and subsurface water balance changes due to burn severity. Vadose Zone Journal 17 (1), 1–13.

Atkinson, G., 1984. Erosion damage following bushfires. Journal of Soil Conservation Service N.S.W. 40, 4–9.

Bachmann, J., van der Ploeg, R.R., 2002. A review on recent developments in soil water retention theory: Interfacial tension and temperature effects. Journal of Plant Nutrition and Soil Science 165, 468–478.

Backlund, P., Janetos, A., Schimel, D., 2008. The Effects of Climate Change on Agriculture, Land Resources, Water Resources, and Biodiversity in the United States. Final Report, Synthesis and Assessment Product 4.3, A Report by the U.S. Climate Change Science Program and the Subcommittee on Global Change Research. US Department of Agriculture, Washington, DC.

Badía, D., Martí, C., 2003. Plant ash and heat intensity effects on chemical and physical properties of two contrasting soils. Arid Land Research and Management 17, 23–41.

Balfour, V.N., 2007. The Effects of Forest Fires on Runoff Rates—The Role of Duff Removal and Surface Sealing on Vegetative Ash, Western Montana. MSc thesis. University of Montana, Missoula, MT.

Balfour, V.N., Doerr, S.H., Robichaud, P.R., 2014. The temporal evolution of wildfire ash and implications for post-fire infiltration. International Journal of Wildland Fire 23, 733–745.

Barrett, S.W., 1988. Fire suppression's effects on forest succession within a central Idaho wilderness. Western Journal of Applied Forestry 3, 76–80.

Benavides-Solario, J., MacDonald, L.H., 2005. Measurement and prediction of post-fire erosion at the Hillslope scale, Colorado Front Range. International Journal of Wildland Fire 14, 1–18.

Benda, L., 1990. The influence of debris flows on channels and valley floors in the Oregon Coast Range, USA. Earth Surface Processes and Landforms 15, 457–466. https://doi.org/10.1002/esp.3290150508.

Benda, L., Dunne, T., 1997. Stochastic forcing of sediment supply to channel networks from landsliding and debris flow. Water Resources Research 33, 2849–2863.

Benda, L., Miller, D., Bigelow, P., Andras, K., 2003. Effects of post-wildfire erosion on channel environments, Boise River, Idaho. Forest Ecology and Management 178, 105–119.

Bernard, D., 2007. Estimation of Inundation Areas of Post-Wildfire Debris Flows. MSc thesis. Colorado School of Mines, Golden, CO.

Beschta, R.L., 1990. Effects of fire on water quantity and quality. In: Walsad, J.S., Radosevich, S.R., Sandberg, D.V. (Eds.), Natural and Prescribed Fire in the Pacific Northwest Forests. Oregon State University Press, Corvallis, OR, pp. 219–231.

Bigio, E., Swetnam, T.W., Baisan, C.H., 2010. A comparison and integration of tree-ring and alluvial records of fire history at the Missionary Ridge fire, Durango, Colorado, USA. Holocene 20, 1047–1061.

UNCORRECTED

Birkeland, P.W., 1984. Soils and Geomorphology. Oxford University Press, New York.
Blackwelder, E., 1927. Fire as an agent in rock weathering. American Journal of Science 35, 134–140.
Blake, W.H., Droppo, I.G., Humphreys, G.S., Doerr, S.H., Shakesby, R.A., Wallbrink, P.J., 2007. Structural characteristics and behavior of fire-modified soil aggregates. Journal of Geophysical Research 112 (F2)https://doi.org/10.1029/2006jf000660.
Booker, F.A., 1998. Landscape and Management Response to Wildfires in California. MSc thesis. University of California, Berkeley, CA.
Brock, R.J., Cannon, S.H., Gartner, J.E., Santi, P.M., Higgins, J.D., Bernard, D.R., 2007. An ordinary storm with an extraordinary response—Mapping the 2003 Old and Grand Prix fire areas in southern California. Geological Society of America Abstracts with Programs 39, 180.
Brogan, D.J., Nelson, P.A., MacDonald, L.H., 2017. Reconstructing extreme post-wildfire floods: A comparison of convective and mesoscale events. Earth Surface Processes and Landforms 42, 2505–2522.
Burroughs, J.E.R., Thomas, B.R., 1977. Declining Root Strength in Douglas-Fir After Felling As a Factor in Slope Stability. US Department of Agriculture Forest Service, 1–27, Research Paper INT-190.
Campbell, R.E., Bakers, M.B., Folliott, P.F., Larsen, F.R., Avery, C.C., 1977. Wildfire Effects on Ponderosa Pine Ecosystems—An Arizona Case Study. USDA Forest Service, Rocky Mountain Forest and Range Experimental Station, Fort Collins, CO, Research Paper RM-191.
Campbell, G., Jungbauer Jr., J., Bidlake, W., Hungerford, R., 1994. Predicting the effect of temperature on soil thermal conductivity. Soil Science 158, 307–313.
Campbell, G., Jungbauer Jr., J., Bristow, K.L., Hungerford, R., 1995. Soil temperature and water content beneath a surface fire. Soil Science 159, 363–374.
Campo, J., Andreu, V., Gimeno-Garcia, E., Gonzalez, O., Rubio, J.L., 2006. Occurrence of soil erosion after repeated experimental fires in a Mediterranean environment. Geomorphology 82, 376–387.
Cannon, S.H., 2001. Debris-flow generation from recently burned watersheds. Environmental and Engineering Geoscience 7, 321–341.
Cannon, S.H., Gartner, J.E., 2005. Wildfire-related debris flow from a hazards perspective. In: Jakob, M., Hungr, O. (Eds.), Debris-Flow Hazards and Related Phenomena. Praxis/Springer, Berlin/Heidelberg, pp. 363–385.
Cannon SH, Ellis WL and Godt JW (1998) Evaluation of the landslide potential in Capulin Canyon following the Dome fire, Bandelier National Monument, New Mexico. U.S. Geological Survey Open-file Report 98-42, 17 pp. Cannon, S.H., Bigio, E.R., Mine, E., 2001. A process for fire-related debris-flow initiation, Cerro Grande Fire, New Mexico. Hydrological Processes 15, 3011–3023.
Cannon, S.H., Kirkham, R.M., Parise, M., 2001. Wildfire-related debris-flow initiation processes, Storm King Mountain, Colorado. Geomorphology 39, 171–188.
Cannon, S.H., Gartner, J.E., Parrett, C., Parise, M., 2003. Wildfire-related debris flow generation through episodic progressive sediment bulking processes, western U.S.A. In: Rickenmann, D., Chen, C.L. (Eds.). Proceedings of the 3rd International Conference on Debris-Flow Hazards Mitigation—Mechanics, Prediction, and Assessment, Davos: Switzerland. pp. 71–82.
Cannon, S.H., Gartner, J.E., Wilson, R.C., Bowers, J.C., Laber, J.L., 2008. Storm rainfall conditions for floods and debris flows for recently burned areas in southwestern Colorado and southern California. Geomorphology 96, 250–269.
Cannon SH, Gartner JE, Rupert MG, Michael JA, Staley DB and Worstell BB (2009) Emergency assessment of postfire debris-flow hazards for the 2009 Station Fire, San Gabriel Mountains, Southern California. U.S. Geological Survey Open-File Report 2009-1227, 27 pp.
Cannon, S.H., Gartner, J.E., Rupert, M.G., Michael, J.A., Rea, A.H., Parrett, C., 2010. Predicting the probability and volume of post-wildfire debris flows in the intermountain western United States. Geological Society of America Bulletin 122, 127–144. https://doi.org/10.1130/B26459.1.
Carabella, C., Miccadei, E., Paglia, G., Sciarra, N., 2019. Post-wildfire landslide hazard assessment: The case of the 2017 Montagna del Morrone fire (Central Apennines, Italy). Geosciences 9, 175.
Cerdà, A., 1988. Post-fire dynamics of erosional processes under Mediterranean climatic conditions. Zeitschrift für Geomorphologie 42, 373–398.
Cerdà, A., Doerr, S.H., 2008. The effect of ash and needle cover on surface runoff and erosion in the immediate post-fire period. Catena 74, 256–263.
.
Certini, G., 2005. Effects of fire on properties of forest soils—A review. Oecologia 143, 1–10.
Chandler, C., Williams, D., Trabaud, L., Thomas, P., Cheney, P., 1983. Fire in forestry. In: Forest Fire Behavior and Effects. vol. 1, John Wiley & Sons, New York.
Chong J, Renaud J and Ailsworth E (2004) *Flash Floods Washed Away Lives, Dreams; Families and Friends Gathered at Campsites Christmas Day in the Mountains Had Little or No Warning*. This is their story. Los Angeles Times, January 3, 2004, p. B1.
Clarke, J., 1990. Fire and rock weathering in Spinifex deserts. In: Western Australia Department of Conservation and Land Management, Spinifex Workshop. p. 15.
Clayton, J.L., Megahan, W.F., 1986. Erosional and chemical denudation rates in the southwestern Idaho batholiths. Earth Surface Processes and Landforms 11, 389–400.
Cleveland, G.B., 1973. Fire + rain = mudflows, Big Sur. California Geology 26, 127–135.
Coe, J., Cannon, S., Santi, P., 2008. Introduction to the special issue on debris flows initiated by runoff, erosion, and sediment entrainment in western North America. Geomorphology 96, 247–249. https://doi.org/10.1016/j.geomorph.2007.05.001.
Conedera, M., Peter, L., Marxer, P., Forster, F., Rickenmann, D., Re, L., 2003. Consequences of forest fires on the hydrogeological response of mountain catchments—A case study of the Riale Buffaga, Ticino, Switzerland. Earth Surface Processes and Landforms 28, 117–129.
Crockford, S., Topalidis, S., Richardson, D.P., 1991. Water repellency in a dry sclerophyll forest—Measurements and processes. Hydrological Processes 5, 405–420.
DeBano, L.F., 2000. The role of fire and soil heating on water repellency in wildland environments: A review. Journal of Hydrology 231–232, 195–206.
DeBano, L.F., Savage, S.M., Hamilton, D.A., 1976. The transfer of heat and hydrophobic substances during burning. Soil Science Society of America Proceedings 40, 779–782.
DeBano, L.F., Rice, R.M., Eugene, C.C., 1979. Soil Heating in Chaparral Fires: Effects on Soil Properties, Plant Nutrients, Erosion, and Runoff. Research Paper RP-145. U.S. Forest Service, Berkeley, CA.
DeBano, L.F., Neary, D.G., Folliott, P.F., 1998. Fire's Effects on Ecosystems. John Wiley and Sons, New York.
DeByle, N.V., 1973. Broadcast burning of logging residues and the water repellency of soils. Northwest Science 47, 77–87.
DeGraff, J.V., 1994. The geomorphology of some debris flows in the southern Nevada, California. Geomorphology 10, 231–252.
DeGraff, J.V., 1997. Geologic Investigation of the Pilot Ridge Debris Flow, Groveland Range District, Stanislaus National Forest. FS-6200-7. U.S. Department of Agriculture, Forest Service, 20.
DeGraff, J.V., Cannon, S.H., Gallegos, A.J., 2007. Reducing post-wildfire debris flow risk through the burned area emergency response (BAER) process. In: Schafer, V.R., Schuster, R.L., Turner, A.K. (Eds.). Conference Presentations, 1st North American Landslide Conference, Association of Engineering Geology Special Publication. vol. 23, pp. 1440–1447.
DeGraff, J.V., Shelmerdine, B., Gallegos, A., Annis, D., 2015. Uncertainty associated with evaluating rockfall hazard to roads in burned areas. Environmental and Engineering Geoscience 21, 21–33.
Dekker, L.W., Ritsema, C.J., 1996. Uneven moisture patterns in water repellent soils. Geoderma 70, 87–99.

UNCORRECTED PROOF

Dekker, L.W., Doerr, S.H., Oostindie, K., Ziogas, A.K., Ritsema, C.J., 2001. Actual water repellency and critical soil water content in a dune sand. Soil Science of America Journal 65, 1667–1675.
Dekker, L.W., Oostindie, K., Ritsema, C.J., 2005. Exponential increase of publications related to soil water repellency. Australian Journal of Soil Research 43, 403–441.
DeLong, S.B., Youberg, A.M., DeLong, W.M., Murphy, B.P., 2018. Post-wildfire landscape change and erosional processes from repeat terrestrial Lidar in a steep headwater catchment, Chiricahua Mountains, Arizona, USA. Geomorphology 300, 13–30.
Desilets, S.L., Nijssen, B., Ekwurzel, B., Ferré, T.P., 2007. Post-wildfire changes in suspended sediment rating curves: Sabino Canyon, Arizona. Hydrological Processes 21, 1413–1423.
Díaz-Fierros, F., Benito, E., Soto, B., 1994. Action of forest fires on vegetation cover and soil erodibility. In: Sala, M., Rubio, J.L. (Eds.), Soil Erosion and Degradation as a Consequence of Forest Fires. vol. 163, Geoforma Ediciones, Logroño, pp. 1–176.
DiBiase, R.A., Lamb, M.P., 2013. Vegetation and wildfire controls on sediment yield in bedrock landscapes. Geophysical Research Letters 40, 1093–1097.
Doehring, D.O., 1969. The effect of fire on geomorphic processes in the San Gabriel Mountains, California. Contributions to Geology 7, 43–65.
Doerr, S.H., Moody, J.A., 2004. Hydrological impacts of soil water repellency—On spatial and temporal uncertainties. Hydrological Processes 18, 829–832.
Doerr, S.H., Thomas, A.D., 2000. The role of soil moisture in controlling water repellency—New evidence from forest soils in Portugal. Journal of Hydrology 231–232, 134–147.
Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., 1996. Soil hydrophobicity variations with depth and particle size fraction in burned and unburned Eucalyptus globulus and Pinus pinaster forest terrain in the Agueda Basin, Portugal. Catena 27, 25–48.
Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., 2000. Soil water repellency—Its causes, characteristics and hydro-geomorphological significance. Earth-Science Reviews 51, 33–65.
Doerr, S.H., Ferreira, A.J.D., Walsh, R.P.D., Shakesby, R.A., Leighton-Boyce, G., Coelho, C.O.A., 2003. Soil water repellency as a potential parameter in rainfall-runoff modeling—Experimental evidence at point to catchment scales from Portugal. Hydrological Processes 17, 363–377.
Doerr, S.H., Blake, W.H., Humphreys, G.S., Shakesby, R.A., Stagnitti, F., Vuurens, S.H., Wallbrink, P., 2004. Heating effects on water repellency in Australian eucalypt forest soils and their value in estimating wildfire soil temperatures. International Journal of Wildland Fire 13, 157–163.
Doerr, S.H., Shakesby, R.A., MacDonald, L.H., 2009. Soil water repellency—A key factor in post-fire erosion. In: Cerdà, A., Robichaud, P.R. (Eds.), Restoration Strategies After Forest Fires. Science Publishers, Enfield, NH, pp. 197–223.
Dorn, R.I., 2003. Boulder weathering and erosion associated with a wildfire, Sierra Ancha Mountains, Arizona. Geomorphology 55, 155–171.
Dragovich, D., 1993. Fire-accelerated boulder weathering in the Pilbara, Western Australia. Zeitschrift für Geomorphogie N.F. 37, 295–307.
Dunne, T., 1978. Field studies of hillslope flow processes. In: Kirkby, M.J. (Ed.), Hillslope Hydrology. John Wiley, New York, pp. 227–293.
Durgin, P.B., 1985. Burning changes the erodibility of forest soils. Journal of Soil and Water Conservation 40, 299–301.
Dyrness, C.T., 1976. Effect of Wildfire on Soil Wettability in the High Cascades of Oregon. USDA Forest Service, Pacific Southwest Forest and Range Experiment Station, 444–447, Research Paper PNW-202.
Eaton, B.C., Moore, R.D., Giles, T.R., 2010. Forest fire, bank strength and channel instability: The 'unusual' response of Fishtrap Creek, British Columbia. Earth Surface Processes and Landforms 35, 1167–1183. https://doi.org/10.1002/esp.1946.
Ebel, B.A., 2012. Wildfire impacts on soil-water retention in the Colorado Front Range, United States. Water Resources Research 48 (12), W12515.
Ebel, B.A., Martin, D.A., 2017. Meta-analysis of field-saturated hydraulic conductivity recovery following wildland fire: Applications for hydrologic model parameterization and resilience assessment. Hydrological Processes 31, 3682–3696.
Ebel, B.A., Moody, J.A., 2013. Rethinking infiltration in wildfire-affected soils. Hydrological Processes 27, 1510–1514.
Ebel, B.A., Moody, J.A., 2017. Synthesis of soil-hydraulic properties and infiltration timescales in wildfire-affected soils. Hydrological Processes 31, 324–340.
Ebel, B.A., Moody, J.A., Martin, D.A., 2012. Hydrologic conditions controlling runoff generation immediately after wildfire. Water Resources Research 48, W03529https://doi.org/10.1029/2011WR011470.
Ebel, B.A., Rengers, F.K., Tucker, G.E., 2016. Observed and simulated hydrologic response for a first-order catchment during extreme rainfall 3 years after wildfire disturbance. Water Resources Research 52, 9367–9389.
Ebel, B.A., Koch, J.C., Walvoord, M.A., 2019. Soil physical, hydraulic, and thermal properties in interior Alaska, USA: Implications for hydrologic response to thawing permafrost conditions. Water Resources Research 55 (5), 4427–4447.
Eccleston, D.T., 2008. Post-Fire Channel Change in Two Small Watersheds in the Colorado Front Range. PhD thesis. Colorado State University, Fort Collins, CO.
Ellett, N.G., Pierce, J.L., Glenn, N.F., 2019. Partitioned by process: Measuring post-fire debris-flow and rill erosion with Structure from Motion photogrammetry. Earth Surface Processes and Landforms 44, 3128–3146.
Elliot, W.J., Liebenow, A.M., Laflen, J.M., Kohl, K.D., 1989. A Compendium of Soil Erodibility Data From WEPP Cropland Field Erodibility Experiments 1987 & 1999. US Department of Agriculture, Agricultural Research Service, National Soil Erosion Research Laboratory NSERL Report Number 3.
Elliott, J.J., Smith, M.E., Friedel, M.J., Stevens, M.R., Bossong, C.R., Litke, D.W., Parker, R.S., Costello, C., Wagner, J., Char, S.J., Bauer, M.A., Wilds, S.R., 2004. Analysis and Mapping of Post-Fire Hydrologic Hazards for the 2002 Hayman, Coal Seam, and Missionary Ridge Wildfires, Colorado. U.S. Geological Survey, 104, Scientific Investigations Report 2004-5300.
Etiegni, L., Campbell, A.G., 1991. Physical and chemical characteristics of wood ash. Bioresource Technology 37, 173–178.
Fahy, R.F., LeBlanc, P.R., 2000. Report on 1999 firefighter fatalities. NFPA Journal 46–61.
Fitch, E.P., Meyer, G.A., 2016. Temporal and spatial climatic controls on Holocene fire-related erosion and sedimentation, Jemez Mountains, New Mexico. Quaternary Research 85, 75–86.
Fitzpatrick, R., 1980. Effect of Forest and Grass Burning on Mineralogical Transformations in Some Soils of Natal. Soil Irrigation Research Institute, 80, Report 952.
Florsheim, J.L., Keller, E.A., Best, D.W., 1991. Fluvial sediment transport following chaparral wildfires, Ventura County, southern California. Geological Society of America Bulletin 103, 504–511.
Florsheim, J.L., Chin, A., O'Hirok, L.S., Storesund, R., 2016. Short-term post-wildfire dry-ravel processes in a chaparral fluvial system. Geomorphology 252, 32–39.
.
Fox, D.M., 2011. Evaluation of the efficiency of some sediment trapping methods after a Mediterranean forest fire. Journal of Environmental Management 92, 258–265.
Frechette, J.D., Meyer, G.A., 2009. Holocene fire-related alluvial-fan deposition and climate in ponderosa pine and mixed-conifer forests, Sacramento Mountains, New Mexico. Holocene 19, 639–651. https://doi.org/10.1177/0959683609104031.
Gabet, E.J., 2003. Post-fire thin debris flows: Sediment transport and numerical modeling. Earth Surface Processes and Landforms 28, 1341–1348.
Gabet, E.J., 2014. Fire increases dust production from chaparral soils. Geomorphology 217, 182–192.
Gabet, E.J., Bookter, A., 2008. A morphometric analysis of gullies scoured by post-fire progressively bulked debris flows in southwest Montana, USA. Geomorphology 96, 298–309.

UNCORRECTED PROOF

Gabet, E.J., Sternberg, P., 2008. The effects of vegetative ash on infiltration capacity, sediment transport and the generation of progressively bulked debris flows. Geomorphology 101, 666–673.
Gartner, J.E., Cannon, S.H., Bigio, E.R., Davis, N.K., Parrett, C., Pierce, K.L., Rupert, M.G., Thurston, B.L., Trebish, M.J., Garcia, S.P., Rea, A.H., 2005. Compilation of Data Relating to the Erosive Response of 606 Recently Burned Basins in the Western US. U.S. Geological Survey, Open File Report 2005-1218.
Gartner, J.E., Cannon, S.H., Santi, P.M., Dewolfe, V.G., 2008. Empirical models to predict the volumes of debris flows generated by recently burned basins in the western U.S. Geomorphology 96, 339–354.
Gartner, J.E., Cannon, S.H., Santi, P.M., 2014. Empirical models for predicting volumes of sediment deposited by debris flows and sediment-laden floods in the transverse ranges of southern California. Engineering Geology 176, 45–56.
Gehring, E., Conedera, M., Maringer, J., Giadrossich, F., Guastini, E., Schwarz, M., 2019. Shallow landslide disposition in burnt European beech (Fagus sylvatica L.) forests. Scientific Reports 9, 1–17.
Germanoski, D., 2002. The importance of event sequencing in the geomorphic impact of wildfire in the central Great Basin. Geological Society of America Abstracts with Programs 34, 143.
Giovannini, G., 1994. The effect of fire on soil quality. In: Sala, M., Rubio, J.L. (Eds.), Soil Erosion as a Consequence of Forest Fires. Geoforma Ediciones, Logroño, pp. 15–27.
Giovannini, G., Lucchesi, S., Giachettti, M., 1988. Effect of heating on some physical and chemical parameters related to soil aggregation and erodibility. Soil Science 146, 255–261.
Goforth, B.R., Graham, R.C., Hubbert, K.R., Zanner, W., Minnich, R.A., 2005. Spatial distribution and properties of ash and thermally altered soils after high-severity forest fire, southern California. International Journal of Wildland Fire 14, 434–454.
Goudie, A.S., Allison, R.J., McLaren, S.J., 1992. The relations between modulus of elasticity and temperature in the context of the experimental simulation of rock weathering by fire. Earth Surface Processes and Landforms 17, 605–615.
Grissino-Mayer, H.D., Romme, W.H., Floyd, L.M., Hanna, D.D., 2004. Climatic and human influences on fire regimes of the southwestern San Juan Mountains, Colorado, USA. Ecology 85, 1708–1724.
Guerrero, C., Mataix-Solera, J., García-Orenes, F., Gómez, I., Navarro-Pedreño, J., 2001. Different patterns of aggregate stability in burned and restored soils. Arid Land Resources Management 15, 163–171.
Hanshaw, M.N., Schmidt, K.M., Stock, J.D., 2009. The role of vegetation canopy removal on post-fire debris-flow rainfall intensity-duration thresholds. Geological Society Abstracts With Programs 41, 495.
Helvey, J.A., 1980. Effects of a north central Washington wildfire on runoff and sediment production. Water Resources Bulletin 16, 627–634.
Hill, J.S., Douglas, T.J., Korte, D.M., Scheip, C.M., Wooten, R.M., Palmer, J.M., 2020. Debris Flows triggered by August 24, 2019 Storm in the Nantahala Gorge, Western North Carolina; Did the Underlying Bedrock and the 2016 Wildfires Increase Landslide Susceptibility?. Geological Society of America Abstracts with Programs 52 (2)https://doi.org/10.1130/abs/2020SE-344730.
Horton, R.E., 1945. Erosional development of streams and their drainage basins; hydrophysical approach to quantitative morphology. Geological Society of America Bulletin 56, 275–370.
Hubbert, K.R., Wohlgemuth, P.M., Preisler, H.K., Graham, R.C., 2002. Effects of prescribed burning on selected soil properties in steep chaparral, southern California. Geological Society of America Abstracts with Programs 34, 206.
Huffman, E.L., MacDonald, L.H., Stednick, J.D., 2001. Strength and persistence of fire-induced soil hydrophobicity under ponderosa and lodgepole pine, Colorado Front Range. Hydrological Processes 15, 2877–2892.
Humphreys, G.S., Shakeby, R.A., Doerr, S.H., Blake, W.H., Wallbrink, P., Hart, D.M., 2003. Some effects of fire on the regolith. In: Roach, I.C. (Ed.), Advances in Regolith. CRC, LEME, Canberra, Australia, pp. 216–220.
Hungr, O., McDougall, S., Bovis, M., 2005. Entrainment of material by debris flows. In: Jakob, M., Hungr, O. (Eds.), Debris-Flow Hazards and Related Phenomena. Praxis/Springer, Berlin, pp. 135–158.
Hyde, K., Woods, S.W., Donahue, J., 2007. Predicting gully rejuvenation after wildfire using remotely sensed burn severity data. Geomorphology 86, 496–511.
Hyde, K.D., Wilcox, A.C., Jencso, K., Woods, S., 2014. Effects of vegetation disturbance by fire on channel initiation thresholds. Geomorphology 214, 84–96.
Istanbulluoglu, E.D.G., Tarbotton, R.T., Pack, R.T., Luce, C.H., 2003. A sediment transport model for incision of gullies on steep topography. Water Resources Research 39, 6–13.
Iverson, R.M., 1997. The physics of debris flow. Reviews in Geophysics 35, 245–296.
Iverson, R.M., Schilling, S.P., Vallance, J.W., 1998. Objective delineation of lahar-inundation hazard zones. Geological Society of America Bulletin 110, 972–984.
Jackson, M., Roering, J.J., 2009. Post-fire geomorphic response in steep, forested landscapes: Oregon Coast Range, USA. Quaternary Science Reviews 29, 1131–1146. https://doi.org/10.1016/j.quascirev.2008.05.003.
Johnson, A.M., 1984. Debris flow. In: Brunsden, D., Prior, D.B. (Eds.), Slope Instability. John Wiley and Sons, New York, pp. 257–361, with contributions by Rodine, J.R..
Jordan, P., 2016. Post-wildfire debris flows in southern British Columbia, Canada. International Journal of Wildland Fire 25, 322–336.
Jordan, P., Covert, S.A., 2009. Debris flows and floods during the 2003 wildfires in Southern British Columbia. Environmental & Engineering Geoscience 15, 217–234.
Jordán, A., Zavala, L.M., Mataix-Solera, J., Nava, A.L., Alanís, N., 2011. Effect of fire severity on water repellency and aggregate stability on Mexican volcanic soils. Catena 84, 136–147.
Kean, J.W., Staley, D.M., Cannon, S.H., 2011. In situ measurements of post-fire debris flows in southern California: Comparisons of the timing and magnitude of 24 debris-flow events with rainfall and soil moisture conditions. Journal of Geophysical Research - Earth Surface 116, F04019.
Kean, J.W., McCoy, S.W., Tucker, G.E., Staley, D.M., Coe, J.A., 2013. Runoff-generated debris flows: Observations and modeling of surge initiation, magnitude, and frequency. Journal of Geophysical Research - Earth Surface 118, 2190–2207. https://doi.org/10.1002/jgrf.20148.
Kean, J.W., McGuire, L., Rengers, F.K., Smith, J.B., Staley, D.M., 2016. Amplification of postwildfire peak flow by debris. Geophysical Research Letters 43, 8545–8553.
Kean, J.W., Staley, D.M., Lancaster, J.T., Rengers, F.K., Swanson, B.J., Coe, J.A., Hernandez, J.L., Sigman, A.J., Allstadt, K.E., Lindsay, D.N., 2019. Inundation, flow dynamics, and damage in the 9 January 2018 Montecito debris-flow event, California, USA: Opportunities and challenges for post-wildfire risk assessment. Geosphere 15 (4), 1140–1163.
Keeley, J.E., 2009. Fire intensity, fire severity and burn severity—A brief review and suggested usage. International Journal of Wildland Fire 18, 116–126.
Keller, E.A., Valentine, D.W., Gibbs, D.R., 1997. Hydrological response of small watersheds following the southern California Painted Cave fire on June 1990. Hydrological Processes 11, 401–414.
Kendrick, K.J., Partin, C.A., Graham, R.C., 2016. Granitic boulder erosion caused by Chaparral wildfire: Implications for cosmogenic radionuclide dating of bedrock surfaces. Journal of Geology 124, 529–539.

UNCORRECTED PROOF

Ketterings, Q.M., Bigham, J.M., Laperche, V., 2000. Changes in soil mineralogy and texture caused by slash-and-burn fires in Sumatra, Indonesia. Soil Science Society of America Journal 64, 1108–1117.
Kinner, D.A., Moody, J.A., 2008. Infiltration and Runoff Measurements on Steep Burned Hillslopes Using a Rainfall Simulator With Variable Rain Intensities. U.S. Geological Survey, 64, Scientific Investigations Report 2007-5211.
Kinoshita, A.M., Hogue, T.S., 2011. Spatial and temporal controls on post-fire hydrologic recovery in Southern California watersheds. Catena 87, 240–252.
Kinoshita, A.M., Hogue, T.S., 2015. Increased dry season water yield in burned watersheds in southern California. Environmental Research Letters 10, 014003.
Kirchner, J.W., Finkel, R.C., Riebe, C.S., Granger, D.E., Clayton, J.L., King, J.G., Megahan, W.F., 2001. Mountain erosion over 10 yr, 10 k.y., and 10 m.y. time scales. Geology 20, 591–594.
Klock, G.O., Helvey, J.D., 1976. Debris flows following wildfire in north central Washington. In: Proceedings of the 3rd Federal Inter-Agency Sedimentation Conference, Denver, CO. pp. 91–98.
Kotok, E.I., Kraebel, C.J., 1935. Discussion of Flood and erosion control problems and their solution. American Society of Civil Engineers Transactions 101, 1350–1355.
Lainas, S., Sabatakakis, N., Koukis, G., 2016. Rainfall thresholds for possible landslide initiation in wildfire-affected areas of western Greece. Bulletin of Engineering Geology and the Environment 75, 883–896.
Lamb, M.P., Scheingross, J.S., Amidon, W.H., Swanson, E., Limaye, A., 2011. A model for fire-induced sediment yield by dry ravel in steep landscapes. Journal of Geophysical Research 116 (F3), F03006.
Lane, P.N., Sheridan, G.J., Noske, P.J., 2006. Changes in sediment loads and discharge from small mountain catchments following wildfire in south eastern Australia. Journal of Hydrology 331, 495–510.
Langhans, C., Lane, P.N., Nyman, P., Noske, P.J., Cawson, J.G., Oono, A., Sheridan, G.J., 2016. Scale-dependency of effective hydraulic conductivity on fire-affected hillslopes. Water Resources Research 52, 5041–5055.
Langhans, C., Nyman, P., Noske, P.J., Van der Sant, R.E., Lane, P.N., Sheridan, G.J., 2017. Post-fire hillslope debris flows: Evidence of a distinct erosion process. Geomorphology 295, 55–75.
Larsen, I.J., 2003. From the Rim to the River—The Geomorphology of Debris Flows in the Green River Canyons of Dinosaur National Monument, Colorado and Utah. MSc thesis. Utah State University, Logan, UT, 196.
Larsen, I.J., MacDonald, L.H., Brown, E., Rough, D., Welsh, M.J., Pietraszek, J.H., Libohova, Z., Benavides-Solorio, J., Schaffrath, K., 2009. Causes of post-fire runoff and erosion—The roles of soil water repellency, cover, and soil sealing. Soil Science of America Journal 73, 1393–1407. https://doi.org/10.2136/sssaj2007.0432.
Lavé, J., Burbank, D., 2004. Denudation processes and rates in the transverse ranges, southern California—Erosional response of a transitional landscape to external and anthropogenic forcing. Journal of Geophysical Research 109, F01006https://doi.org/10.1029/2003JF000023.
Legleiter, C.J., Lawrence, R.L., Fonstad, M.A., Marcus, W.A., Aspinall, R., 2003. Fluvial response a decade after wildfire in the northern Yellowstone ecosystem: A spatially explicit analysis. Geomorphology 54, 119–136.
Leonard, J.M., Magana, H.A., Bangert, R.K., Neary, D.G., Montgomery, W.L., 2017. Fire and floods: The recovery of headwater stream systems following high-severity wildfire. Fire Ecology 13, 62–84.
Lipovsky, P.S., Coates, J., Lewkowicz, A.G., Trochim, E., 2006. Active-layer detachments following the summer 2004 forest fires near Dawson City, Yukon. In: Emond, D.S., Bradshaw, G.D., Lewis, L.L., Weston, L.H. (Eds.), Yukon Exploration and Geology 2005. Yukon Geological Survey, Whitehorse, YT, pp. 175–194.
Littell J.S., McKenzie D., Peterson D.L., Westerling A.L., Climate and wildfire area burned in western U.S. ecoprovinces, 1916–2003, *Ecological Applications* **19**, 2009, 1003–1021. Loaiciga, H.A., Pederos, D., Roberts, D., 2001. Wildfire-streamflow interactions in a chaparral watershed. Advances in Environmental Research 5, 295–305.
Long, C.J., Whitlock, C., Bartlein, P.J., Millspaugh, S.H., 1998. A 9000-year fire history from the Oregon Coast Range based on a high-resolution charcoal study. Canadian Journal of Forest Research 28, 774–787.
MacDonald, L.H., Huffman, E.L., 2004. Post-fire soil water repellency—Persistence and soil moisture thresholds. Soil Science Society of America Journal 68, 1729–1734.
Mallik, A.U., Gimingham, C.H., Rahman, A.A., 1984. Ecological effects of heather burning, I. Water infiltration, moisture retention and porosity of surface soils. Journal of Ecology 72, 767–776.
Malmon, D.V., Reneau, S.L., Katzman, D., Lavine, A., Lyman, J., 2007. Suspended sediment transport in an ephemeral stream following wildfire. Journal of Geophysical Research - Earth Surface 112, F02006.
Martin, Y.E., 2007. Wildfire disturbance and shallow landsliding in coastal British Columbia over millennial time scales: A numerical modelling study. Catena 69, 206–219.
Martin, D.A., Moody, J.A., 2001. Comparison of soil infiltration rates in burned and unburned mountainous watersheds. Hydrological Processes 15, 2893–2903.
Massman, W., 2012. Modeling soil heating and moisture transport under extreme conditions: Forest fires and slash pile burns. Water Resources Research 48, W10548.
Massman, W., Frank, J., 2005. Effect of a controlled burn on the thermophysical properties of a dry soil using a new model of soil heat flow and a new high temperature heat flux sensor. International Journal of Wildland Fire 13, 427–442.
Mataix-Solera, J., Doerr, S.H., 2004. Hydrophobicity and aggregate stability in calcareous topsoils from fire-affected pine forests in south eastern Spain. Geoderma 118, 77–88.
Mataix-Solera, J., Guerrero, C., García-Orenes, F., Bárcenas, G.M., Torres, P.M., 2009. Forest fire effects on soil microbiology. In: Cerdà, A., Robichaud, P.R. (Eds.), Restoration Strategies after Forest Fires. vol. 5, Science Publishers, Enfield, NH, pp. 133–176.
Mataix-Solera, J., Cerdà, A., Arcenegui, V., Jordán, A., Zavala, L.M., 2011. Fire effects on soil aggregation: A review. Earth-Science Reviews 109, 44–60.
May, C.L., 2002. Debris flow occurrence in the immediate post-fire and inter-fire time periods, and the associated effects on channel aggradation in the Oregon Coast Range, USA. Geological Society of America Abstracts with Programs 34, 143.
May, C.L., Gresswell, R.E., 2003. Processes and rates of sediment and wood accumulation in headwater streams of the Oregon Coast Range, USA. Earth Surface Processes and Landforms 28, 409–424.
McDonald, G.N., Giraud, R.E., 2002. Fire-Related Debris Flows East of Santaquin and Spring Lake, Utah County, Utah. Utah Geological Survey, 15, Technical Report 02-09.
McGuire, L.A., Kean, J.W., Staley, D.M., Rengers, F.K., Wasklewicz, T.A., 2016. Constraining the relative importance of raindrop-and flow-driven sediment transport mechanisms in postwildfire environments and implications for recovery time scales. Journal of Geophysical Research - Earth Surface 121, 2211–2237.
McGuire, L.A., Rengers, F.K., Kean, J.W., Staley, D.M., 2017. Debris flow initiation by runoff in a recently burned basin: Is grain-by-grain sediment bulking or en masse failure to blame?. Geophysical Research Letters 44, 7310–7319.

UNCORRECTED PROOF

McGuire, L.A., Rengers, F.K., Kean, J.W., Staley, D.M., Mirus, B.B., 2018. Incorporating spatially heterogeneous infiltration capacity into hydrologic models with applications for simulating post-wildfire debris flow initiation. Hydrological Processes 32, 1173–1187.

Meeuwig, R.O., 1971. Infiltration and Water Repellency in Granitic Soils. USDA Forest Service, Intermountain Forest and Range Experimental Station, Ogden, UT, 20, Forest Service Research Paper INT-111.

Megahan, W.F., 1983. Hydrologic effects of clearcutting and wildfire on steep granitic slopes in Idaho. Water Resources Research 19, 811–819.

Megahan, W.F., Day, N.F., Bliss, T.M., 1978. Landslide occurrence in the western and central Northern Rocky Mountain physiographic province in Idaho. In: Proceedings 5th of the North American Soils Conference, Fort Collins, CO. pp. 116–139.

Meyer, G.A., Wells, S.G., 1997. Fire-related sedimentation events on alluvial fans, Yellowstone National Park, USA. Journal of Sedimentary Research 67, 776–791.

Meyer, G., Wells, S.G., Balling Jr., R., Jull, A.J.T., 1992. Response of alluvial systems to fire and climate change in Yellowstone National Park. Nature 357, 147–150.

Meyer, G.A., Wells, S.G., Jull, A.J.T., 1995. Fire and alluvial chronology in Yellowstone National Park: Climatic and intrinsic controls on Holocene geomorphic processes. Geological Society of America Bulletin 107, 1211–1230.

Meyer, G.A., Pierce, J.L., Wood, S.H., Jull, A.J.T., 2001. Fire, storms, and erosional events in the Idaho batholiths. Hydrological Processes 15, 3025–3038.

Meyer, G.A., Pierce, J., New, J., Frechette, J., 2006. Episodic erosion forced by fire and climate in the western United States. Geological Society of America Abstracts with Programs 38, 174.

Meyer, G.A., Pierce, J., Frechette, J., New, J., Nelson, N., Persico, L., 2008. Medieval-period droughts, fires, floods and geomorphic change in interior western USA mountains. Geological Society of America Abstracts with Programs 40, 228.

Michaelides, R.J., Schaefer, K., Zebker, H.A., Parsekian, A., Liu, L., Chen, J., Natali, S., Ludwig, S., Schaefer, S.R., 2019. Inference of the impact of wildfire on permafrost and active layer thickness in a discontinuous permafrost region using the remotely sensed active layer thickness (ReSALT) algorithm. Environmental Research Letters 14, 035007.

Mills, A.J., Fey, M.V., 2004. Frequent fires intensify soil crusting—Physiochemical feedback in the pedoderm on long-term burn experiments in South Africa. Geoderma 121, 423–437.

Minsley, B.J., Pastick, N.J., Wylie, B.K., Brown, D.R., Andy Kass, M., 2016. Evidence for nonuniform permafrost degradation after fire in boreal landscapes. Journal of Geophysical Research - Earth Surface 121 (2), 320–335.

Mirus, B.B., Loague, K., 2013. How runoff begins (and ends): Characterizing hydrologic response at the catchment scale. Water Resources Research 49, 2987–3006. https://doi.org/10.1002/wrcr.20218.

Molina, M.J., García-Fayos, P., Sanroque, P., 1994. Short-term changes on aggregate stability and organic matter content after forest fires in a calcareous soil in Valencia (Spain). In: Sala, M., Rubio, J.L. (Eds.), Soil Erosion and Degradation as a Consequence of Forest Fires. Geoforma Ediciones, Logroño, Spain, pp. 43–52.

Moody, J.A., Ebel, B.A., 2012. Hyper-dry conditions provide new insights into extreme floods after wildfire. Catena 93, 58–63.

Moody, J.A., Martin, D.A., 2001. Initial hydrologic and geomorphic response following a wildfire in the Colorado Front Range. Earth Surface Processes and Landforms 26, 1049–1070.

Moody, J.A., Martin, D.A., Geomorphic consequences of wildfire, *Geological Society of America Abstracts with Programs* **34**, 2002, 143.

Moody, J.A., Martin, D.A., 2007. Forest fire effects on geomorphic processes. In: Cerdà, A., Robichaud, P.R. (Eds.), Restoration Strategies After Forest Fires. vol. 5, Science Publishers, Enfield, NH, pp. 41–79.

Moody, J.A., Martin, D.A., 2009. Synthesis of sediment yields after Wildland fire in different rainfall regimes in the western Unites States. International Journal of Wildland Fire 18, 96–115.

Moody, J.A., Nyman, P., 2013. Variations in Soil Detachment Rates After Wildfire as a Function of Soil Depth, Flow Properties and Root Properties. U.S. Geological Survey, 40, Scientific Investigations Report 2012-5233.

Moody, J.A., Smith, J.D., Ragan, B., 2005. Critical shear stress for erosion of cohesive soils subjected to temperatures typical of wildfires. Journal of Geophysical Research - Earth Surface 110, https://doi.org/10.1029/2004JF000141.

.

Moody, J.A., Shakesby, R.A., Robichaud, P.R., Cannon, S.H., Martin, D.A., 2013. Current research issues related to post-wildfire runoff and erosion processes. Earth-Science Reviews 122, 10–37. https://doi.org/10.1016/j.earscirev.2013.03.004.

Moody, J.A., Ebel, B.A., Nyman, P., Martin, D.A., Stoof, C., McKinley, R., 2016. Relations between soil hydraulic properties and burn severity. International Journal of Wildland Fire 25, 279–293.

Morton, D.M., 1989. Distribution and frequency of storm-generated soil slips on burned and unburned slopes, San Timoteo Badlands, Southern California. In: Sadler, P.M., Morton, D.M. (Eds.), Landslides in a Semi-Arid Environment with Emphasis on the Inland Valleys of Southern California. vol. 2, Publications of the Inland Geological Society, pp. 279–284.

Muir, R.A., Bordy, E.M., Prevec, R., 2015. Lower Cretaceous deposit reveals first evidence of a post-wildfire debris flow in the Kirkwood Formation, Algoa Basin, Eastern Cape, South Africa. Cretaceous Research 56, 161–179.

Murphy, B.P., Yocom, L.L., Belmont, P., 2018. Beyond the 1984 perspective: Narrow focus on modern wildfire trends underestimates future risks to water security. Earth's Future 6, 1492–1497.

Murphy, B.P., Czuba, J.A., Belmont, P., 2019. Post-wildfire sediment cascades: A modeling framework linking debris flow generation and network-scale sediment routing. Earth Surface Processes and Landforms 44, 2126–2140.

Neary, D., Gottfried, G., 2002. Fires and floods: post-fire watershed responses. In: Viegas, D. (Ed.). Forest Fire Research and Wildland Fire Safety, Proceedings of IV International Conference on Forest Fire Research, 2002 Wildland Fire Safety Summit. Millpress Science Publishers, Luso, Coimbra, Portugal.

Neary, D.G., Klopatek, C.C., DeBano, L.F., Folliott, P.F., 1999. Fire effects on belowground sustainability: A review and synthesis. Forest Ecology and Management 122, 51–71.

Neary, D.G., Ryan, K.C., DeBano, L.F., 2005. Wildland Fire in Ecosystems—Effects of Fire on Soil and Water. vol. 4, USDA Forest Service, Rocky Mountain Research Station, Ogden, Utah, General Technical Report RMRS-GTR-42.

Noske, P.J., Nyman, P., Lane, P.N., Sheridan, G.J., 2016. Effects of aridity in controlling the magnitude of runoff and erosion after wildfire. Water Resources Research 52, 4338–4357.

Nyman, P., Sheridan, G., Lane, P.N., 2010. Synergistic effects of water repellency and macropore flow on the hydraulic conductivity of a burned forest soil, south-east Australia. Hydrological Processes 24, 2871–2887.

Nyman, P., Sheridan, G.J., Smith, H.G., Lane, P.J., 2011. Evidence of debris flow occurrence after wildfire in upland catchments of south-east Australia. Geomorphology 125, 383–401.

Nyman, P., Sheridan, G.J., Moody, J.A., Smith, H.G., Noske, P.J., Lane, P.N., 2013. Sediment availability on burned hillslopes. Journal of Geophysical Research - Earth Surface 118, 2451–2467.

Nyman, P., Sheridan, G.J., Smith, H.G., Lane, P.N., 2014. Modeling the effects of surface storage, macropore flow and water repellency on infiltration after wildfire. Journal of Hydrology 513, 301–313.

UNCORRECTED PROOF

Nyman, P., Box, W.A., Stout, J.C., Sheridan, G.J., Keesstra, S.D., Lane, P.N., Langhans, C., 2019. Debris-flow dominated sediment transport through a channel network after wildfire. Earth Surface Processes and Landforms 45 (5), 1155–1167.
O'Brien, J.S., 1993. FLO-2D Two-dimensional Flood Routing System, Version 2001.06. FLO-2D Software, Inc., Nutrioso, AZ, CD-ROM.
Oakley, N.S., Lancaster, J.T., Kaplan, M.L., Ralph, F.M., 2017. Synoptic conditions associated with cool season post-fire debris flows in the transverse ranges of southern California. Natural Hazards 88, 327–354.
Oakley, N.S., Lancaster, J.T., Hatchett, B.J., Stock, J., Ralph, F.M., Roj, S., Lukashov, S., 2018. A 22-year climatology of cool season hourly precipitation thresholds conducive to shallow landslides in California. Earth Interactions 22, 1–35.
Ollier, C.D., Ash, J.E., 1983. Fire and rock breakdown. Zeitschrift für Geomorphologie 27, 363–374.
Onda, Y., Dietrich, W.E., Booker, F., 2008. Evolution of overland flow after a severe wildfire, Point Reyes, California. Catena 72, 13–20.
Parrett, C., 1987. Fire-Related Debris Flows in the Beaver Creek Drainage, Lewis and Clark County, Montana. U.S. Geological Survey, 57–67, Water-Supply Paper 2330.
Parrett, C., Cannon, S.H., Pierce, K.L., 2003. Wildfire-Related Floods and Debris Flows in Montana in 2001. U.S. Geological Survey, Water-Resources Investigations Report 03-4319.
Parson, A., Robichaud, P.R., Lewis, S.A., Napper, C., Clark, J.T., 2010. Field Guide for Mapping Post-Fire Burn Severity. USDA Forest Service, General Technical Report RMRS-GTR-243. Rocky Mountain Research Station, Fort Collins, CO, 49.
Pelletier, J.D., Orem, C.A., 2014. How do sediment yields from post-wildfire debris-laden flows depend on terrain slope, soil burn severity class, and drainage basin area? Insights from airborne-LiDAR change detection. Earth Surface Processes and Landforms 39, 1822–1832.
Pereira, P., Cerdà, A., Úbeda, X., Mataix-Solera, J., Martin, D., Jordán, A., Burguet, M., 2013. Spatial models for monitoring the spatio-temporal evolution of ashes after fire—A case study of a burnt grassland in Lithuania. Solid Earth 4 (1), 153–165.
Poon, P.K., Kinoshita, A.M., 2018. Spatial and temporal evapotranspiration trends after wildfire in semi-arid landscapes. Journal of Hydrology 559, 71–83.
Prochaska, A.B., Santi, P.M., Higgins, J.D., Cannon, S.H., 2008. Debris-flow runout predictions based on the average channel slope (ACS). Engineering Geology 98, 29–40.
Prochaska, A.B., Santi, P.M., Higgins, J.D., 2008. Debris basin and deflection berm design for fire-related debris-flow mitigation. Environmental and Engineering Geoscience 14, 297–313.
Prosser, I.P., 1990. Fire, humans and denudation at Wangrah Creek, southern Tablelands, NSW. Australian Geographical Studies 28, 77–95.
Prosser, I.P., Williams, L., 1998. The effect of wildfire on runoff and erosion in native Eucalyptus forest. Hydrological Processes 12, 251–265.
Puri, A.N., Asghar, A.G., 1940. Physical characteristics of soils, VII. Effect of ignition. Soil Science 49, 369–378.
Ragan, B.W., 2002. Effect of wildfire on rill erosion. Geological Society of America Abstracts with Programs 34, 206.
Ravi, S., D'Odorico, P., Zobeck, T., Over, T., Collins, S.L., 2006. Enhancement of wind erosion by fire-induced water repellency. Water Resources Research 42 (11), W11422.
Reneau, S.L., Katzman, D., Kuyumjian, G.A., Lavine, A., Malmon, D.V., 2007. Sediment delivery after a wildfire. Geology 35, 151–154.
Rengers, F., McGuire, L., Kean, J.W., Staley, D.M., Hobley, D., 2016. Model simulations of flood and debris flow timing in steep catchments after wildfire. Water Resources Research 52, 6041–6061.
Rengers, F., Tucker, G.E., Moody, J.A., Ebel, B.A., 2016. Illuminating Wildfire Erosion and Deposition Patterns with Repeat Terrestrial Lidar. Journal of Geophysical Research - Earth Surface 121, 588–608. https://doi.org/10.1002/2015JF003600.
Rengers, F.K., Pagonis, V., Mahan, S.A., 2017. Can thermoluminescence be used to determine soil heating from a wildfire?. Radiation Measurements 107, 119–127.
Rengers, F.K., McGuire, L.A., Ebel, B.A., Tucker, G.E., 2018. The evolution of a colluvial hollow to a fluvial channel with periodic steps following two transformational disturbances: A wildfire and a historic flood. Geomorphology 309, 121–130.
Rengers, F.K., McGuire, L.A., Kean, J.W., Staley, D.M., Youberg, A.M., 2019. Progress in simplifying hydrologic model parameterization for broad applications to post-wildfire flooding and debris-flow hazards. Earth Surface Processes and Landforms 44, 3078–3092.
Rice, R.M., 1982. Sedimentation in the chaparral: How do you handle unusual events?. In: Swanson, F.J., Janda, R.J., Dunne, T. (Eds.). Proceedings of Workshop on Sediment Budgets and Routing in Forested Drainage Basins. USDA Forest Service, Pacific Northwest Forest and Range Experiment Station, Portland, Oregon, General Technical Report PNW-141.
Rice, R.M., Foggin III, G.T., 1971. Effect high intensity storms on soil slippage on mountainous watersheds in Southern California. Water Resources Research 7, 1485–1496.
Riebe, C., 2008. Wildfires, floods and sediment delivery at the wilderness-urban interface in coastal southern California. In: Teaching Geomorphology in the 21st Century, National Association of Geology Teachers Workshop, http://serc.carleton.edu/NAGTWorkshops/geomorph/vignettes/25470.html, accessed 12/12/08.
Riley, K., Pierce, J., Meyer, G.A., 2015. Vegetative and climatic controls on Holocene wildfire and erosion recorded in alluvial fans of the Middle Fork Salmon River, Idaho. Holocene 25, 857–871.
Robichaud, P.R., Brown, R.E., 1999. What happened after the smoke cleared: Onsite erosion rates after a wildfire in eastern Oregon. In: Olsen, D.S., Potyondy, J.P. (Eds.). Proceedings of the Wildland Hydrology Conference. American Water Resources Association. pp. 419–426.
Robichaud, P.R., Hungerford, R.D., 2000. Water repellency by lab burning of four northern Rocky Mountain forest soils. Journal of Hydrology 231–232, 207–219.
Robichaud, P.R., Miller, S.M., 1999. Spatial interpolation and simulation of post-burn duff thickness after prescribed fire. International Journal of Wildland Fire 9, 137–143.
Robichaud, P.R., Wagenbrenner, J., Brown, R., 2010. Rill erosion in natural and disturbed forests: 1. Measurements. Water Resources Research 46, W10506https://doi.org/10.1029/2009WR008314.
Roering, J.J., Gerber, M., 2005. Fire and the evolution of steep, soil-mantled landscapes. Geology 33, 349–352.
Ryan, K.C., 2002. Dynamic interactions between forest structure and fire behavior in boreal ecosystems. Silva Fennica 36, 13–39.
Ryan, K.C., Noste, N.V., 1985. Evaluating Prescribed Fires. USDA Forest Service Intermountain Forest and Range Experiment Station, 230–238, General Technical Report INT-182.
Sanborn, P., Geertsema, M., Jull, A.J.T., Hawkes, B., 2006. Soil and sedimentary charcoal evidence for Holocene forest fires in an inland temperate rainforest, east-central British Columbia, Canada. Holocene 16, 415–427.
Santi, P.M., MacAulay, B., 2020. Water and sediment supply requirements for post-wildfire debris flows in the western U.S. Environmental and Engineering Geoscience, in review.
Santi, P.M., Morandi, L., 2013. Comparison of debris-flow volumes from burned and unburned areas. Landslides 10, 757–769.
Santi, P.M., deWolfe, V.G., Higgins, J.D., Cannon, S.H., Gartner, J.E., 2008. Sources of debris flow material in burned areas. Geomorphology 96, 310–321. https://doi.org/10.1016/j.geomorph.2007.02.022.
Sass, O., Heel, M., Leistner, I., Stöger, F., Wetzel, K.F., Friedmann, A., 2012. Disturbance, geomorphic processes and recovery of wildfire slopes in North Tyrol. Earth Surface Processes and Landforms 37, 883–894.

UNCORRECTED PROOF

Savage, S.M., Osborn, J., Letey, J., Heton, C., 1972. Substances contributing to fire induced water repellency in soils. Proceedings of the Soil Science Society of America 36, 674–678.
Schmidt, K.M., Roering, J.J., Stock, J.D., Dietrich, W.E., Montgomery, D.R., Schuab, T., 2001. The variability of root cohesion as an influence on shallow landslide susceptibility in the Oregon Coast Range. Canadian Geotechnical Journal 38, 995–1024.
Schmidt, K.M., Stock, J.D., Hanshaw, M.N., 2007. Post-fire sediment flux of the Day fire, California. Eos 88 (52), H43F–1682.
Schmidt, K.M., Hanshaw, M.N., Howle, J.F., Staley, D.M., Stock, J.D., Bawden, G.W., 2011. Hydrologic conditions and terrestrial laser scanning of post-fire debris flows in the San Gabriel Mountains, CA, USA. In: Proceedings, 5th International Conference on Debris-flow Hazards Mitigation: Mechanics, Prediction and Assessment. Padua, Italy.
Schulz, W.H., Coe, J.A., Ellis, W.L., Kibler, J.D., 2006. Preliminary Assessment of Landslides Along the Florida River Downstream From Lemon Reservoir, La Plata County, Colorado. U.S. Geological Survey, 43, Open-File Report 2006–1343.
Schwartz GE, Alexander RB (1995) Soils Data for the Conterminous United States Derived from the NRCS State Soil Geographic (STATSGO) Data Base. U.S. Geological Survey. https://water.usgs.gov/GIS/metadata/usgswrd/XML/ussoils.xml. Accessed 01 July 2020. Scott, R.F., 1963. Principles of Soil Mechanics. Addison-Wesley Publishing Company, Inc, Reading MA.
Scott, K.M., 1971. Origin and Sedimentology of 1969 Debris Flows near Glendora, California. U.S. Geological Survey, C242–C247, Professional Paper 750.
Scott, D.F., Versfeld, D.B., Lesch, W., 1998. Erosion and sediment yield in relation to afforestation and fire in the mountains of the Western Cape Province, South Africa. South African Geographical Journal 80, 52–59.
Selkirk, P.M., Adamson, D., 1981. The effect of fire on Sydney sandstone. In: Stanbury, P. (Ed.), Bushfires: Their Effect on Australian Life and Landscape. Macleay Museum, Sydney, NSW, pp. 24–31.
Shakesby, R.A., Doerr, S.H., 2006. Wildfire as a hydrological and geomorphological agent. Earth-Science Reviews 74, 269–307.
Shakesby, R.A., Coelho, C.O.A., Ferreira, A.D., Terry, J.P., Walsh, R.P.D., 1993. Wildfire impacts on soil erosion and hydrology in wet Mediterranean forest, Portugal. International Journal of Wildland Fire 3, 95–110.
Shaub, S., 2001. Landslides and Wildfire—An Example From the Boise National Forest. MSc thesis. Boise State University, Boise, ID.
Sheridan, G.J., Lane, P.N.J., Noske, P.J., 2007. Quantification of hillslope runoff and erosion processes before and after wildfire in a wet Eucalyptus forest. Journal of Hydrology 343, 12–28.
Shtober-Zisu, N., Tessler, N., Tsatskin, A., Greenbaum, N., 2015. Accelerated weathering of carbonate rocks following the 2010 wildfire on Mount Carmel, Israel. International Journal of Wildland Fire 24, 1154–1167. https://doi.org/10.1071/WF14221.
Sidle, R.C., 1991. A conceptual model of changes in root cohesion in response to vegetation management. Journal of Environmental Quality 20, 43–52.
Sidle, R.C., 1992. A theoretical model of the effects of timber harvesting on slope stability. Water Resources Research 28, 1897–1910.
Sidle, R.C., 2005. Chapter 16: Influence of forest harvesting activities on debris avalanches and flows. In: Jakob, M., Hungr, O. (Eds.), Debris-flow Hazards and Related Phenomena. Praxis/Springer, Berlin, pp. 387–409.
Sidle, R.C., Ochiai, H., 2006. Landslides: Processes, Prediction, and Land Use. Water Resources Monograph, vol. 18, American Geophysical Union.
Singer, M.J., Le Bissonnais, Y., 1998. Importance of surface sealing in the erosion of some soils from a Mediterranean climate. Geomorphology 24, 79–85.
Smith, H.G., Sheridan, G.J., Lane, P.N., Nyman, P., Haydon, S., 2011. Wildfire effects on water quality in forest catchments: A review with implications for water supply. Journal of Hydrology 396, 170–192.
Soler, M., Sala, M., Gallart, F., 1994. Post fire evolution of runoff and erosion during an eighteen month period. In: Sala, M., Rubio, J.L. (Eds.), Soil Erosion and Degradation as a Consequence of Forest Fires. Geoforma Ediciones, Logroño, pp. 149–162.
Spittler, T.E., 1995. Fire and debris flow potential of winter storms. In: Keely, J.E., Scott, T. (Eds.), Brushfires in California Wildlands—Ecology and Resource Management. International Association of Wildland Fire, Fairfield, WA, pp. 113–120.
Spittler, T.E., Wagner, D.L., **51** (3), 1998. Geology and slope stability along Highway 50. California Geology 3–16.
Staley, D.M., Wasklewicz, T.A., Kean, J.W., 2010. Observations of Drainage Network Change in a Recently Burned Watershed Using Terrestrial Laser Scanning. European Geophysical Union Research, Abstracts 12, 2010-4849-1.
Staley, D.M., Kean, J.W., Cannon, S.H., Schmidt, K.M., Laber, J.L., 2013. Objective definition of rainfall intensity–duration thresholds for the initiation of post-fire debris flows in southern California. Landslides 10, 547–562.
Staley, D.M., Wasklewicz, T.A., Kean, J.W., 2014. Characterizing the primary material sources and dominant erosional processes for post-fire debris-flow initiation in a headwater basin using multi-temporal terrestrial laser scanning data. Geomorphology 214, 324–338.
Staley, D.M., Negri, J.A., Kean, J.W., Laber, J.L., Tillery, A.C., Youberg, A.M., 2017. Prediction of spatially explicit rainfall intensity–duration thresholds for post-fire debris-flow generation in the western United States. Geomorphology 278, 149–162.
Staley, D.M., Tillery, A.C., Kean, J.W., McGuire, L.A., Pauling, H.E., Rengers, F.K., Smith, J.B., 2018. Estimating post-fire debris-flow hazards prior to wildfire using a statistical analysis of historical distributions of fire severity from remote sensing data. International Journal of Wildland Fire 27, 595–608.
Steele, R., Arno, S.F., Geier-Hayes, K., 1986. Wildfire patterns change in Central Idaho's ponderosa pine—Douglas fir forest. Western Journal of Applied Forestry 1, 16–18.
Stocks, B.J., Fosberg, M.A., Lynham, T.J., Mearns, L., Wotton, B.M., Yang, Q., Jin, J.Z., Lawrence, K., Hartley, G.R., Mason, J.A., McKenney, D.W., 1998. Climate change and forest fire potential in Russian and Canadian boreal forests. Climatic Change 38, 1–13.
Stoof, C.R., Wesseling, J.G., Ritsema, C.J., 2010. Effects of fire and ash on soil water retention. Geoderma 159, 276–285.
Stoof, C.R., Gevaert, A.I., Baver, C., Hassanpour, B., Morales, V.L., Zhang, W., Martin, D., Giri, S.K., Steenhuis, T.S., 2016. Can pore-clogging by ash explain post-fire runoff?. International Journal of Wildland Fire 25, 294–305.
Swanson, F.J., 1981. Fire and geomorphic processes. In: Mooney, H.A., Bonniksen, T.H., Christensen, N.L., Lotan, J.E., Reiners, W.A. (Eds.), Fire Regimes and Ecosystem Properties. USDA Forest Service, pp. 401–420, General Technical Report WO-26.
Takahashi, T., 1981. Debris flow. Annual Review of Fluid Mechanics 13, 57–77.
Tang, H., McGuire, L.A., Rengers, F.K., Kean, J.W., Staley, D.M., Smith, J.B., 2019. Evolution of debris flow initiation mechanisms and sediment sources during a sequence of post-wildfire rainstorms. Journal of Geophysical Research 124, 1572–1595. https://doi.org/10.1029/2018JF004837.
Tang, H., McGuire, L.A., Rengers, F.K., Kean, J.W., Staley, D.M., Smith, J.B., 2019. Developing and testing physically based triggering thresholds for runoff-generated debris flows. Geophysical Research Letters 46, 8830–8839.
Thode, A.E., Kerschner, J.L., Roby, K., Decker, L.M., Beyers, J.L., 2006. Fire, watershed resources, and aquatic ecosystems. In: Sugihara, N.G., Van Wagtendonk, J.W., Shaffer, K.E., Fites-Kaufman, J., Thode, A.E. (Eds.), Fire in California's Ecosystems. University of California Press, Berkeley, CA, pp. 466–480.
Tiedemann, A.R., Conrad, C.E., Dieterich, J.H., Hornbeck, J.W., Megahan, W.F., Vireck, L.A., Wades, D.D., 1979. Effects of Fire on Water, A State-of-the-Art Review. USDA Forest Service, General Technical Report WO-10.
Tillery, A.C., Rengers, F.K., 2019. Controls on debris-flow initiation on burned and unburned hillslopes during an exceptional rainstorm in southern New Mexico, USA. Earth Surface Processes and Landforms 45 (4), 1051–1066.
Tognacca, C., Bezzola, G.R., 1997. Debris-flow initiation by channel-bed failure. In: Chen, C.L. (Ed.). Debris-Flow Hazards Mitigation: Mechanics, Prediction, and Assessment, Proceedings of the 1st International Conference, San Francisco, CA, August 7–9, American Society of Civil Engineers. pp. 44–53.

UNCORRECTED PROOF

Tognacca, C., Bezzola, G.R., Minor, H.E., 2000. Threshold criterion for debris-flow initiation due to channel bed failure. In: Wieczorek, G.F., Nasser, N.D. (Eds.), Debris-Flow Hazards Mitigation: Mechanics, Prediction, and Assessment. Balkema, Rotterdam, pp. 89–97.
U.S. Geological Survey, 2020. Grizzly Creek Fire Debris-flow Assessment, Retrieved from https://landslides.usgs.gov/hazards/postfire_debrisflow/detail.php?objectid=317.
Ulery, A.L., Graham, R.C., Amrhein, C., 1993. Wood-ash composition and soil pH following intense burning. Soil Science 156, 358–364.
Ulery, A., Graham, R., Bowen, L., 1996. Forest fire effects on soil phyllosilicates in California. Soil Science Society of America Journal 60, 309–315.
Van Wagtendonk, J.W., 2006. Fire as a physical process. In: Sugihara, N.G., Van Wagtendonk, J.W., Shaffer, K.E., Fites-Kaufman, J., Thode, A.E. (Eds.), Fire in California's Ecosystems. University of California Press, Berkeley, CA, pp. 38–57.
VanDine, D.F., Rodman, R.F., Jordan, P., Dupas, J., 2005. Kuskonook Creek, an example of a debris flow analysis. Landslides 2, 257–265.
Vergani, C., Werlen, M., Conedera, M., Cohen, D., Schwarz, M., 2017. Investigation of root reinforcement decay after a forest fire in a Scots pine (Pinus sylvestris) protection forest. Forest Ecology and Management 400, 339–352.
Vierick, L.A., 1982. Effects of fire and firelines on active layer thickness and soil temperature in interior Alaska. In: French, H.M. (Ed.). The Roger J.E. Brown Memorial Volume, Proceedings, Fourth Canadian Permafrost Conference, National Research Council of Canada, Ottawa, ON. pp. 123–135.
Wagenbrenner, J.W., MacDonald, L.H., Rough, D., 2006. Effectiveness of three post-fire treatments in the Colorado Front Range. Hydrological Processes 20, 2989–3006.
Wagenbrenner, J., Robichaud, P., Elliot, W., 2010. Rill erosion in natural and disturbed forests: 2. Modeling approaches. Water Resources Research 46 (10), W10507https://doi.org/10.1029/2009WR008315.
Wall, S., Roering, J., Rengers, F.K., 2020. Runoff-initiated post-fire debris flow Western Cascades, Oregon. Landslides https://doi.org/10.1007/s10346-020-01376-9.
Walsh, R.P.D., Boakes, D.J., Coelho, C.O.A., Ferreira, A.J.D., Shakesby, R.A., Thomas, A.D., 1995. Post-fire land use and management and runoff responses to rainstorms in northern Portugal. In: McGregor, D.F.M., Thompson, D.A. (Eds.), Geomorphology and Land Management in a Changing Environment. John Wiley and Sons, Chichester, pp. 283–308.
Wan, Z., Song, T., 1987. The effect of fine particles on vertical concentration distribution and transport rate of coarse particles. In: White, W.R. (Ed.). Topics in Fluvial Hydraulics—Proceedings 22nd Congress, International Association for Hydraulic Research, Lausanne, Switzerland. pp. 80–85.
Wells WG (1981) Some effects of brushfires on erosion processes in coastal Southern California. In: Davies TRH, Pearce AJ (eds.) *Erosion and Sediment Transport in Pacific Rim Steeplands, Christchurch, New Zealand, International Association of Hydrological Science*, vol. 132, pp 305–342. Wallingford, Oxfordshire, UK: IAHS Press. Wells, W.G., 1987. The effect of fire on the generation of debris flows in southern California. In: Costa, J.E., Wieczorek, G.F. (Eds.), Debris Flows/Avalanches: Process, Recognition, and Mitigation. Geological Survey of America Reviews in Engineering Geology, vol. 7, pp. 105–113.
Wentz, F.J., Ricciardulli, L., Hilburn, K., Mills, C., 2007. How much more rain will global warming bring?. Science 317, 233–235.
Westerling, A.L., Hidalgo, H.G., Cayan, D.R., Swetnam, T.W., 2006. Warming and earlier spring increase western U.S. forest wildfire activity. Science 313, 940–943.
Wohl, E.E., Pearthree, P.P., 1991. Debris flows as geomorphic agents in the Huachuca Mountains of southeastern Arizona. Geomorphology 4, 273–292.
Wohlgemuth, P.M., Hubbert, K., Arbaugh, M.J., 2006. Fire and physical environment interactions. In: Sugihara, N.G., Van Wagtendonk, J.W., Shaffer, K.E., Fites-Kaufman, J., Thode, A.E. (Eds.), Fire in California's Ecosystems. University of California Press, Berkeley, CA, pp. 75–93.
Wondzell, S.M., King, J.G., 2003. Post fire erosional processes in the Pacific Northwest and Rocky Mountains Regions. Forest Ecology and Management 178, 75–87.
Woods, S.W., Balfour, V.N., 2008. The effect of ash on runoff and erosion after a severe forest fire, Montana, USA. International Journal of Wildland Fire 17, 535–548.
Woods, S.W., Balfour, V.N., 2010. The effects of soil texture and ash thickness on the post-fire hydrological response from ash-covered soils. Journal of Hydrology 393, 274–286.
Woods, S.W., Birkas, A., Ahl, R., 2007. Spatial variability of soil hydrophobicity after wildfires in Montana and Colorado. Geomorphology 86, 465–479.
Youberg, A., 2008. Delineating Post-Wildfire Debris Flow Hazards for Pre-fire Mitigation, Pine and Strawberry, Arizona—A FEMA 5% Initiative Study. Arizona Geological Survey, 36, Open-file Report OFR 08-08.
Zhu, J.C., Gantzer, C.J., Anderson, S.H., Peyton, R.L., Alberts, A.A., 2001. Comparison of concentrated-flow detachment equations at low shear stress. Soil and Tillage Research 61, 203–212.

UNCORRECTED PROOF